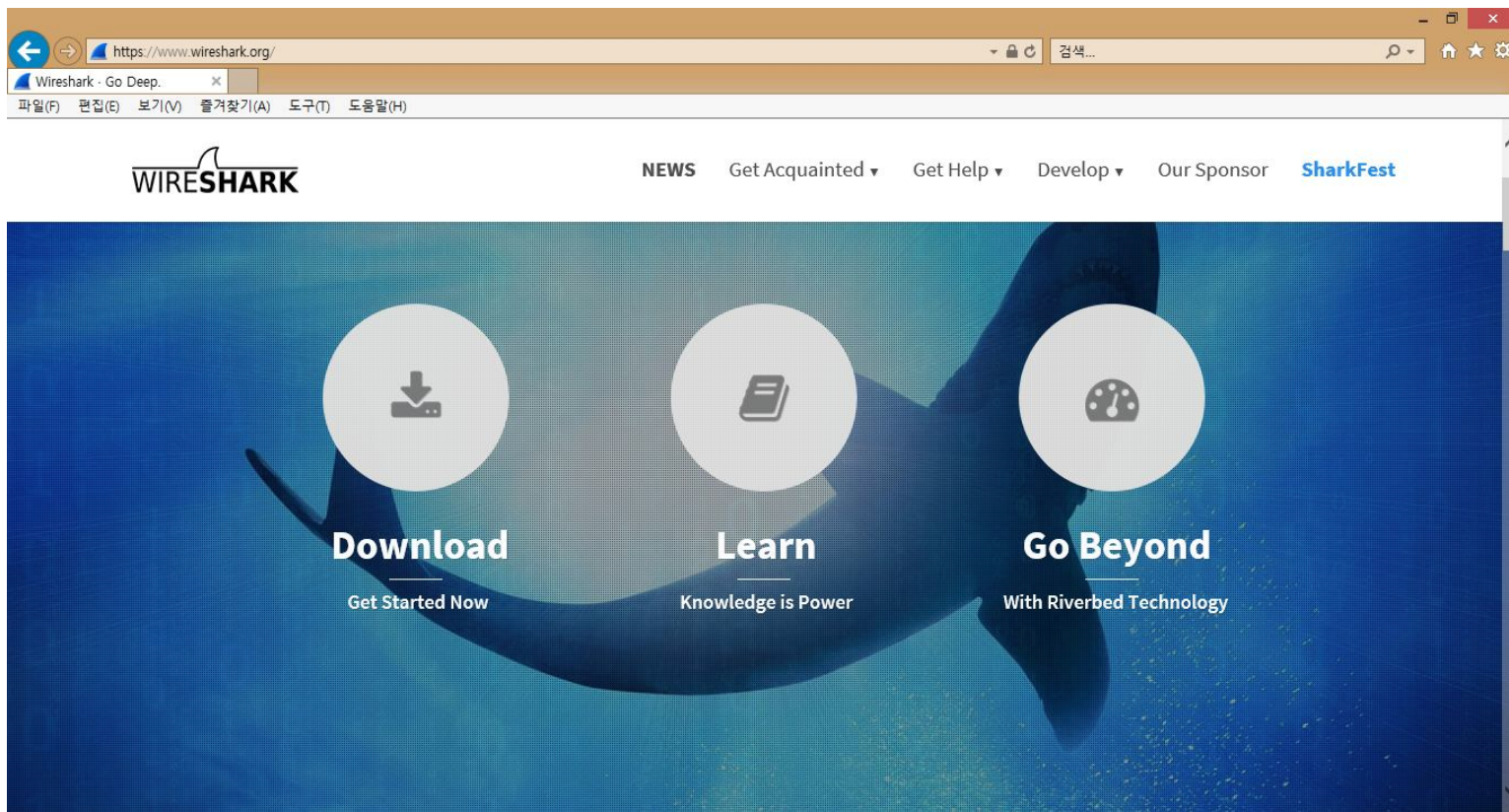


Sniffing 환경구성과 Wireshark

Sniffing 환경구성

- 패킷 스니핑(sniffing) 또는 프로토콜 분석(protocol analysis) 툴
- 네트워크를 통해 전달되는 데이터를 수집하고 해석



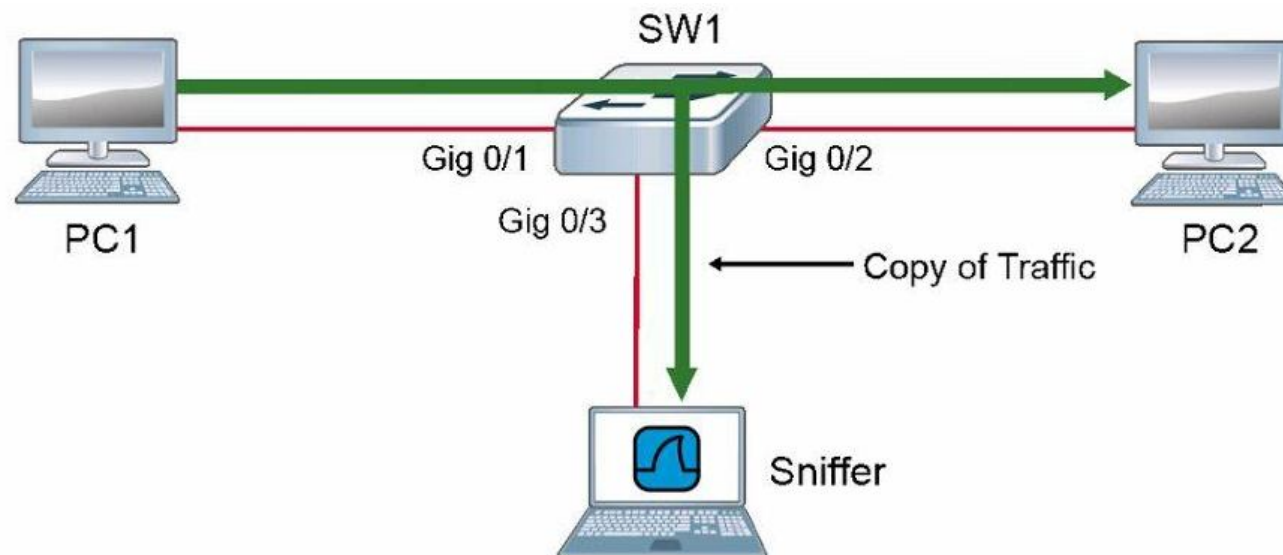
<http://www.wireshark.org>

* 패킷 수집과 패킷 분석

	LINUX	MS-Window
패킷수집	TCPdump	TShark
패킷분석	Wireshark	

무차별모드(Promiscuous mode)

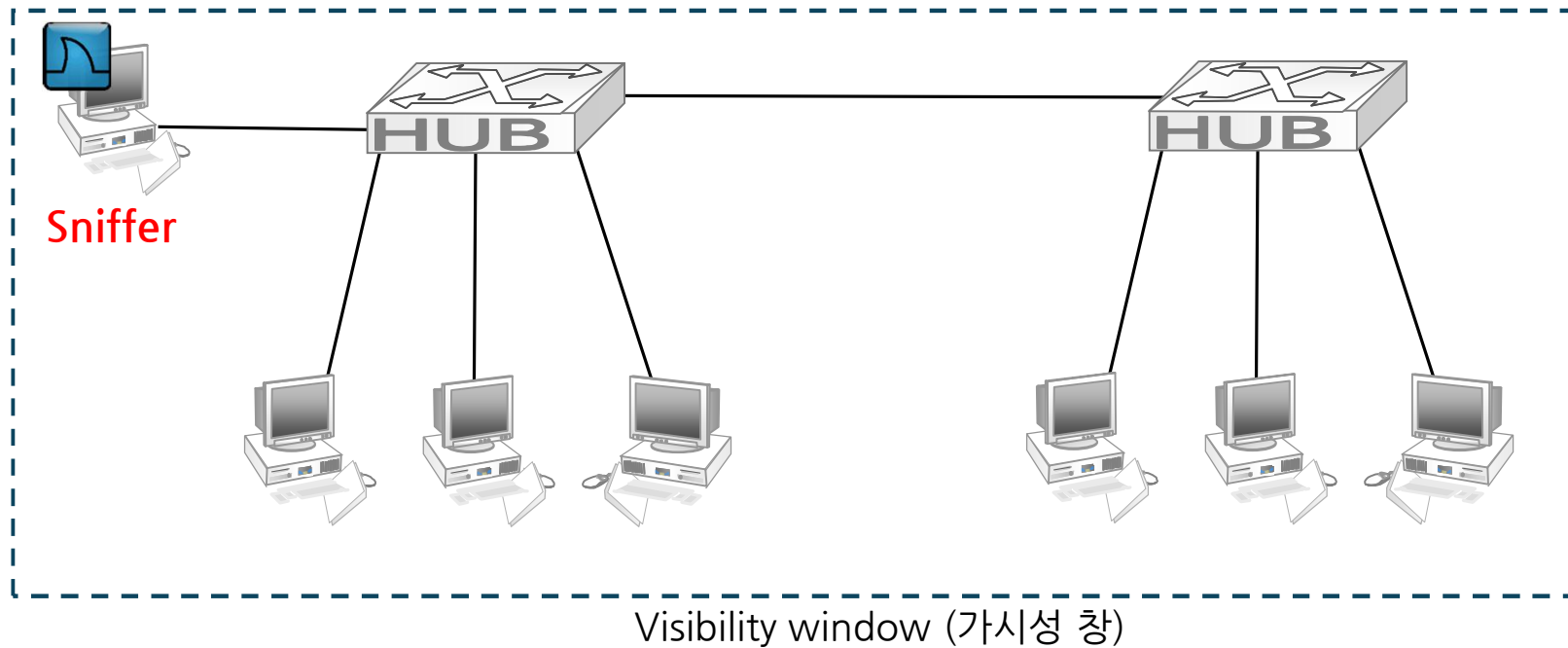
- 데이터 링크 계층과 네트워크 계층의 주소 필터링을 해제한 모드
- 주소 지정에 관계없이 호스트의 프로세서에 모든 패킷을 전달
- 패킷이 CPU에 전달되면 분석을 위해 패킷 스니핑 애플리케이션에 넘겨짐
- 유닉스 또는 리눅스에서는 'ifconfig eth0 promisc'로 설정
- Window에서는 npcap 설치로 무차별 모드 설정



Packet Sniffing 환경

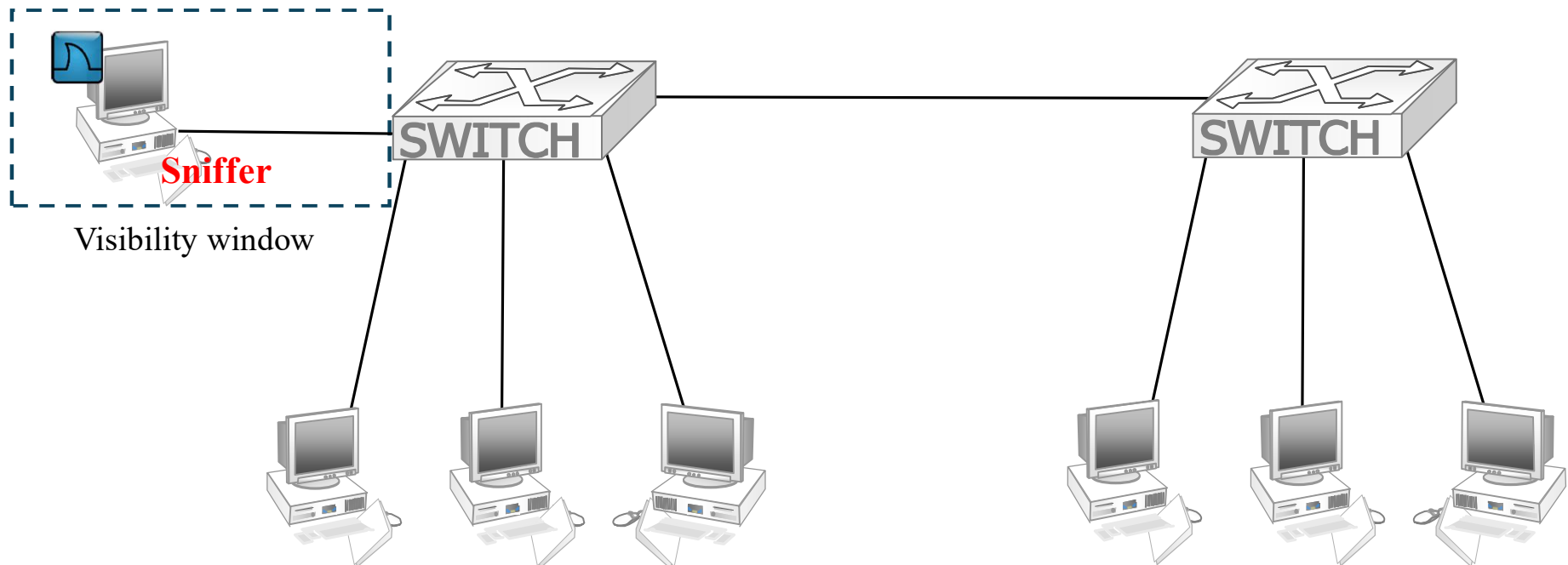
❶ 허브 환경에서의 스니핑

- 허브를 통해 전송되는 트래픽은 해당 허브에 연결된 모든 포트를 통과

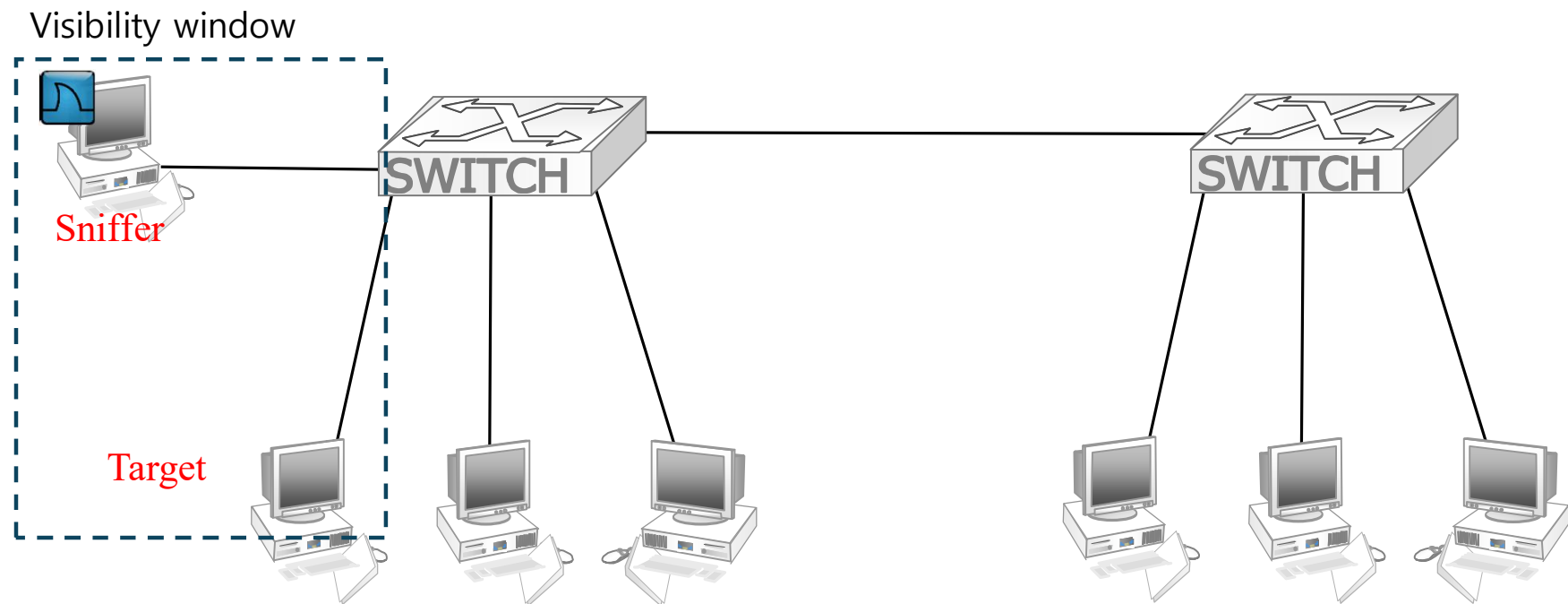


② 스위치 환경에서 스니핑 방법

- 포트 미러링(port mirroring)
- 허빙 아웃(hubbing out)
- 탭 사용 (tapping)



포트 미러링

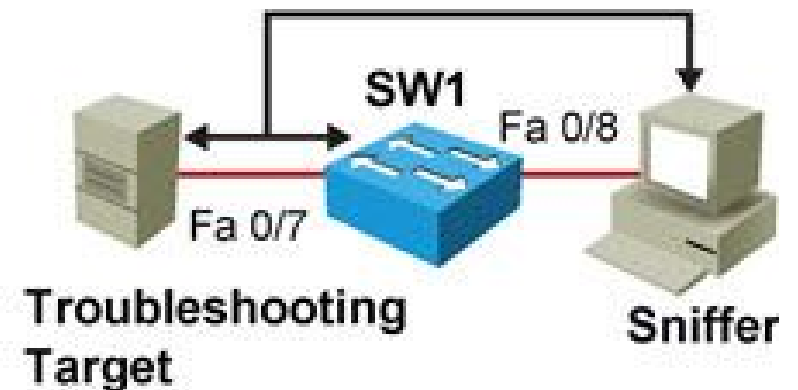


•Switched Port Analyzer (SPAN)

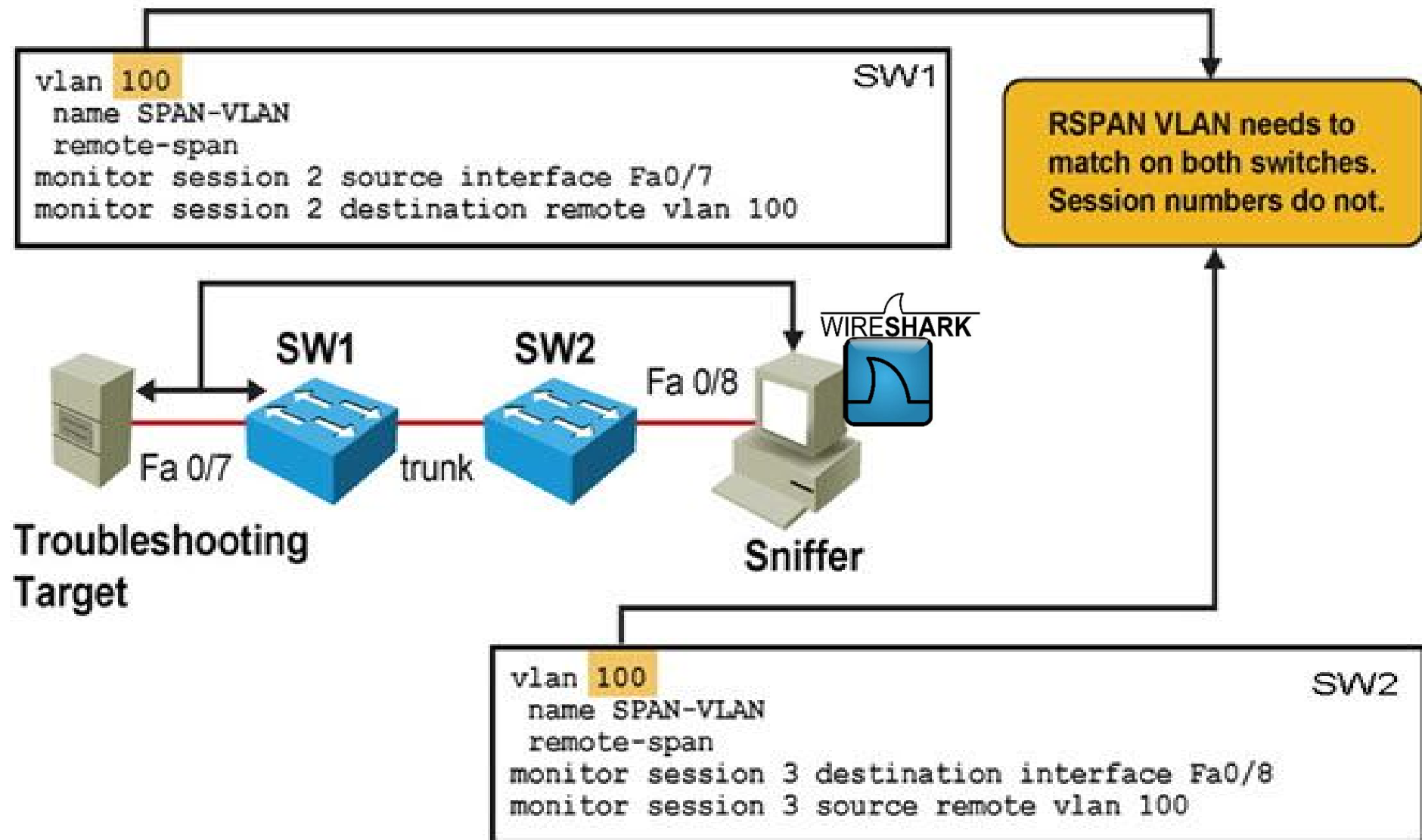
```
monitor session 1 source interface Fa0/7  
monitor session 1 destination interface Fa0/8
```

Sources and destinations that form a single SPAN session are identified by a session number

```
SW1#show monitor  
Session 1  
-----  
Type : Local Session  
Source Ports :  
    Both : Fa0/7  
Destination Ports : Fa0/8  
Encapsulation : Native  
Ingress : Disabled
```

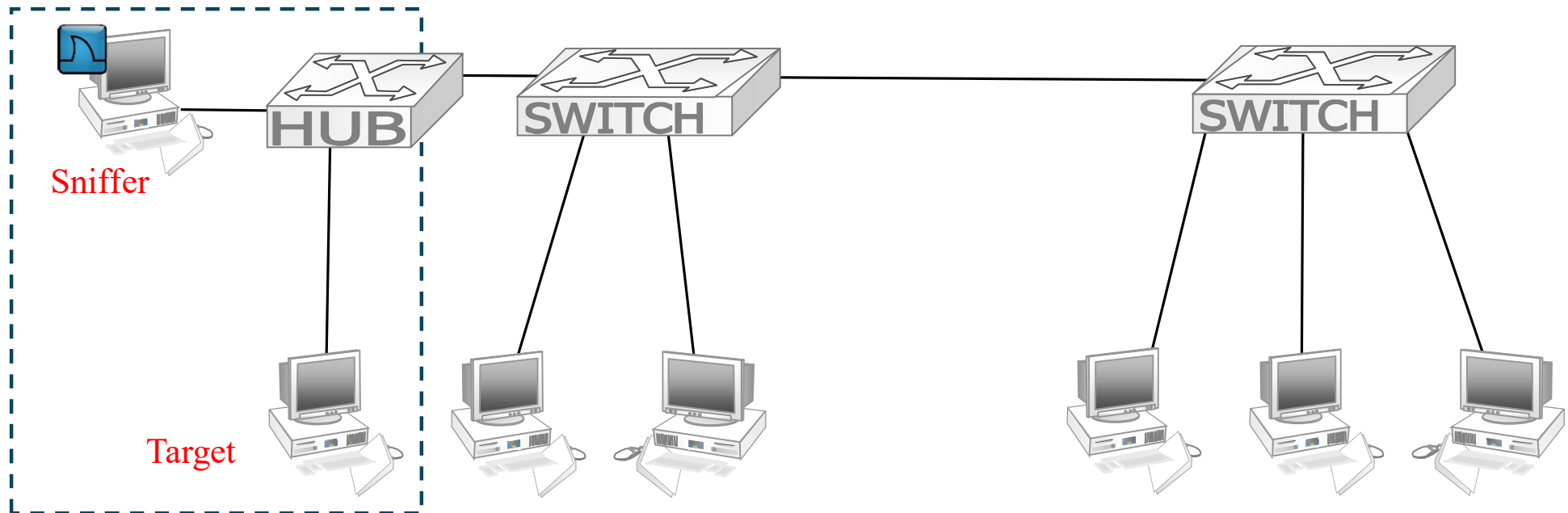


- Remote Switched Port Analyzer (RSPAN)



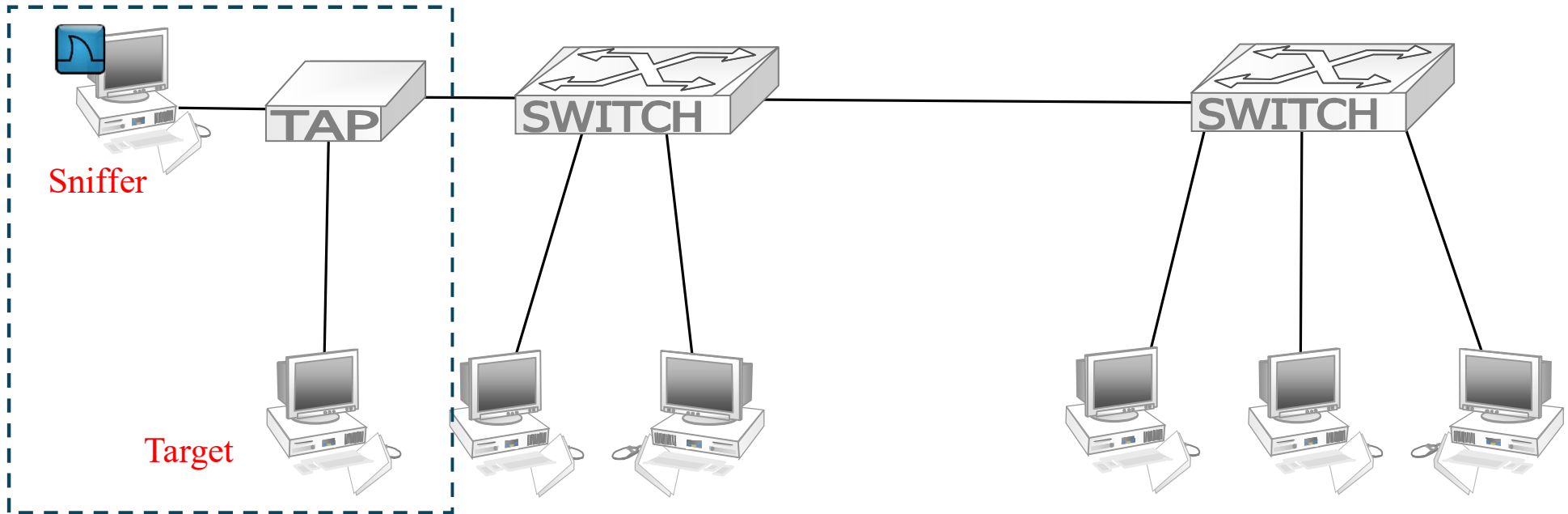
허빙 아웃(hubbing out)

Visibility window



탭 사용 (tapping)

Visibility window

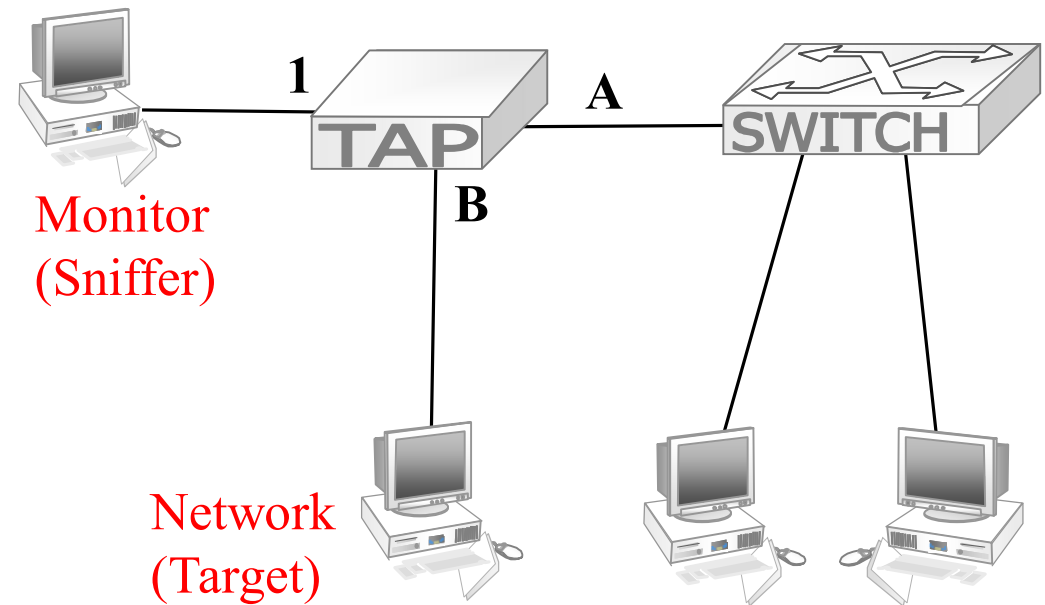
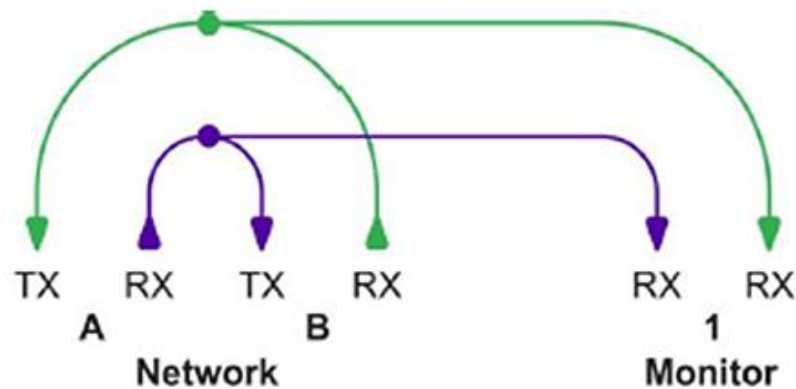


TAP(Test Access Ports)

- 데이터 흐름의 중단 없이, 네트워크에 전혀 영향을 주지 않으면서 트래픽을 모니터링 할 수 있게 해주는 모니터링 장비
- 최근 네트워크 탭은 스위치처럼 12포트, 24포트가 있어 많은 수의 장비를 동시에 사용가능

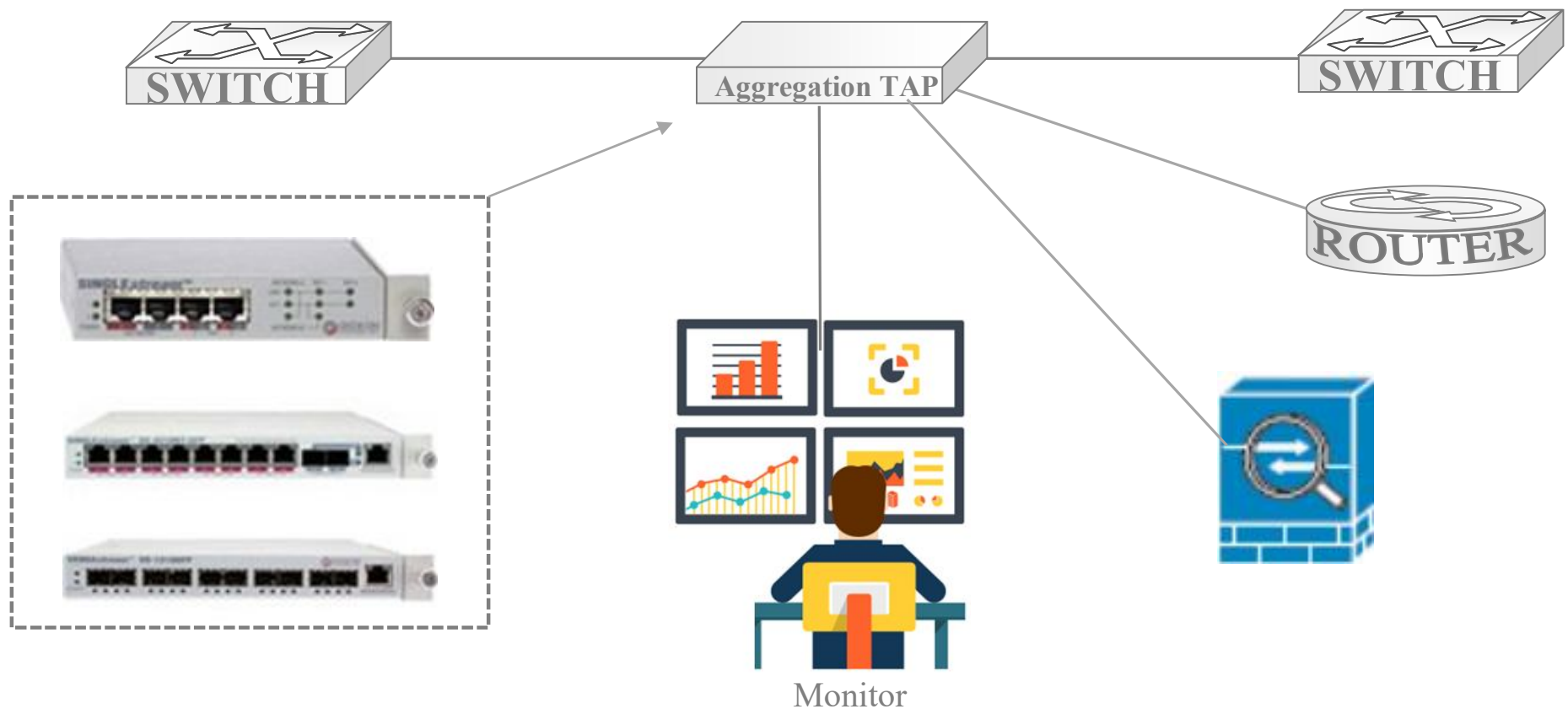
스위치 환경에서의 스니핑 - Aggregated Tap

- 3개의 포트 구성
- 양방향 트래픽을 스니핑하기 위한 물리적 모니터 포트를 하나만 갖고 있음

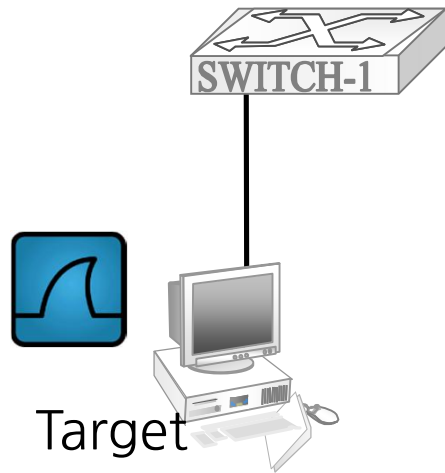


Aggregation TAP

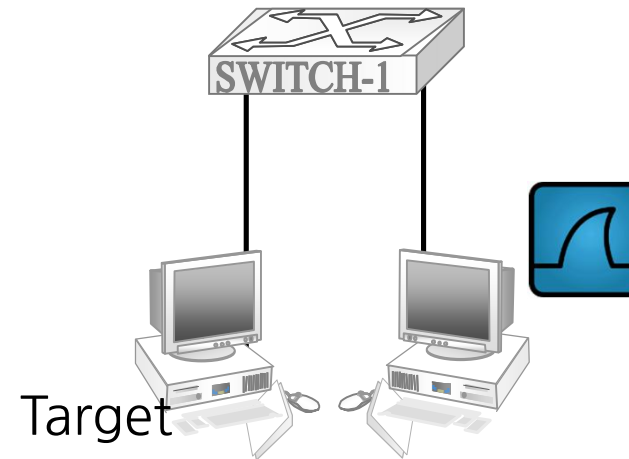
- 1개 이상의 Network Links 또는 Ports에서 수집한 Packet Data들을, 1개의 NIC를 사용하는 Monitor 또는 그 이상의 여러 Monitor 장비로 보내 분석할 수 있게 해 주는 TAP



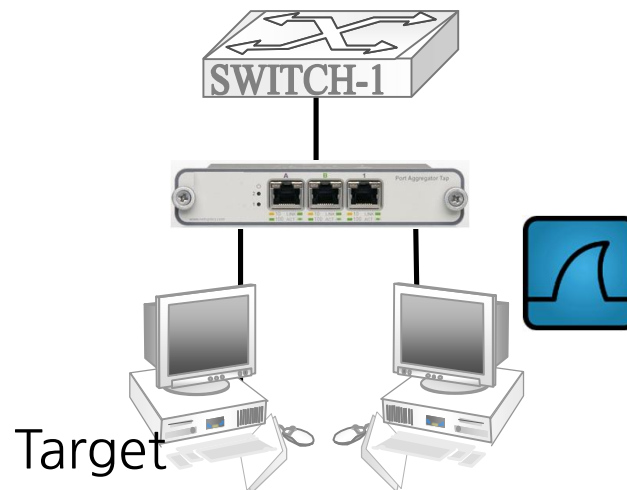
① 지연이 발생한 호스트에서 직접 수집



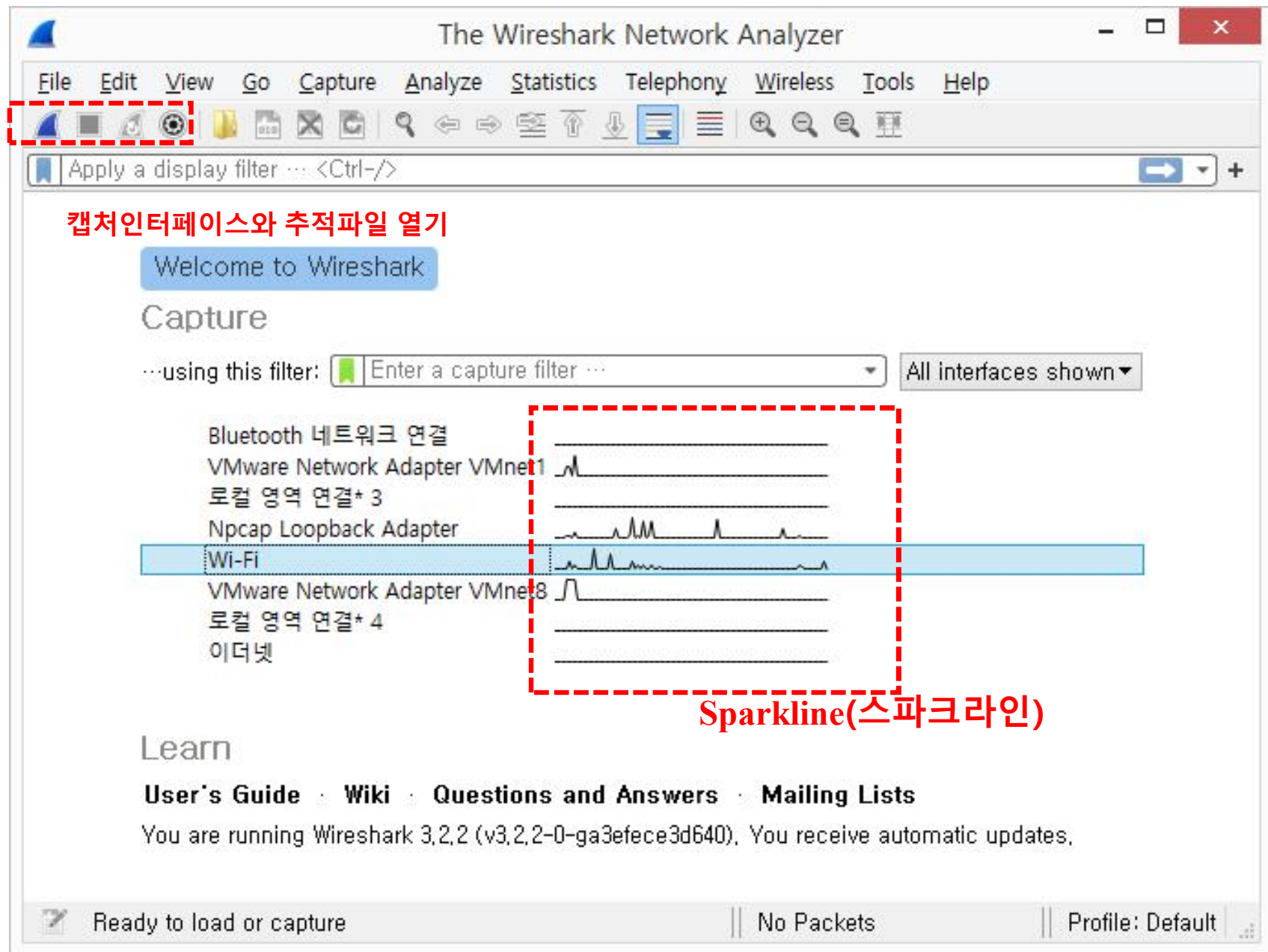
② 호스트의 스위치 포트를 확장(포트미러링)



③ TAP 설정



Wireshark Tool 기능

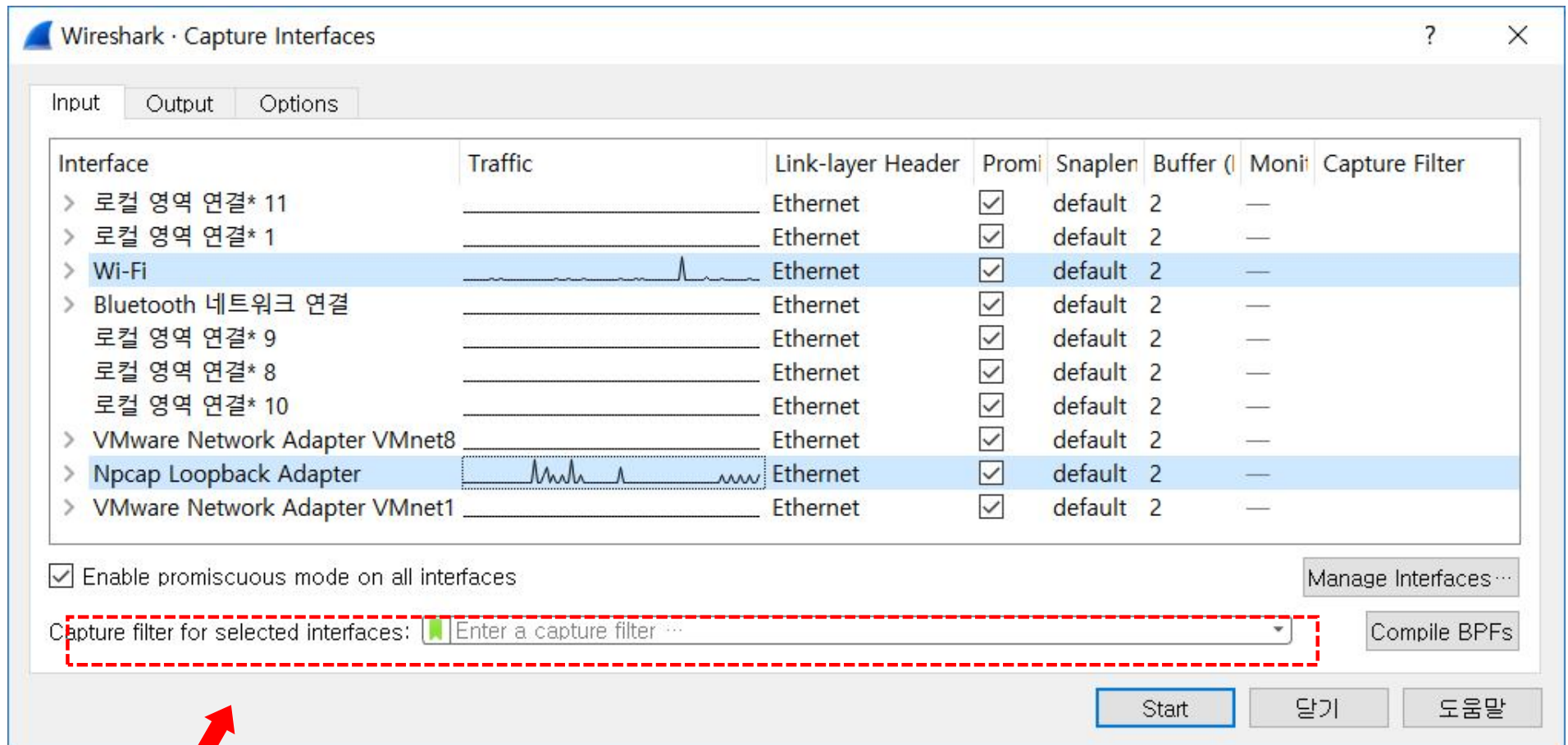


Wireshark Filter

- 캡처 필터(Capture Filter)
 - 패킷이 캡처 될 때 지정
 - 지정된 표현식에 포함/제외된 패킷만 캡처
- 디스플레이 필터(Display Filter)
 - 원하지 않는 패킷을 숨김
 - 지정된 표현식을 기반으로 원하는 패킷을 보기

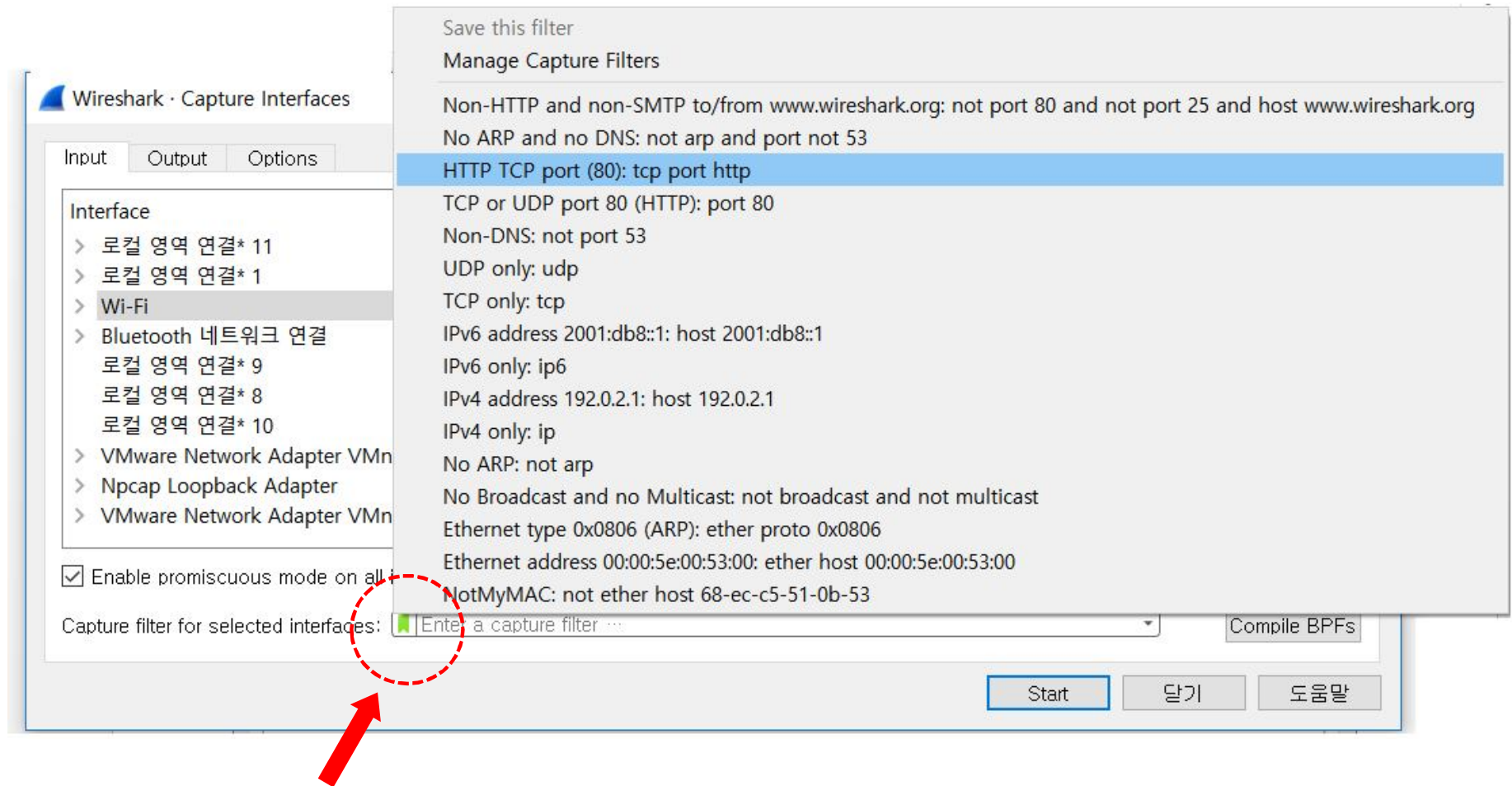
1) Capture Filter

- Capture Option 창에서 수집 필터 적용



Capture Option 창에서 수집 필터 적용

❶ Capture Option > 수집 필터 체크할피 화살표



수집 필터 체크할피 화살표

주소 기반의 트래픽 수집

① 특정 IP 주소에서/로 오는 트래픽 수집

- host 10.3.1.1
- host 2406:da00:ff00::6b16:f02d
- not host 10.3.1.1
- src host 10.3.1.1
- dst host 10.3.1.1
- host 10.3.1.1 or host 10.3.1.2
- host www.espn.com

주소 기반의 트래픽 수집

② IP 주소 범위에서/로 오는 트래픽 수집

- net 10.3.0.0/16
- net 10.3.0.0 mask 255.255.0.0
- ipv6 net 2406:da00:ff00::/64
- not dst net 10.3.0.0/16
- dst net 10.3.0.0/16
- src net 10.3.0.0/16

주소 기반의 트래픽 수집

③ 브로드캐스트 또는 멀티캐스트 트래픽 수집

- ip broadcast
- ip multicast
- dst host ff02::1
- dst host ff02::2

주소 기반의 트래픽 수집

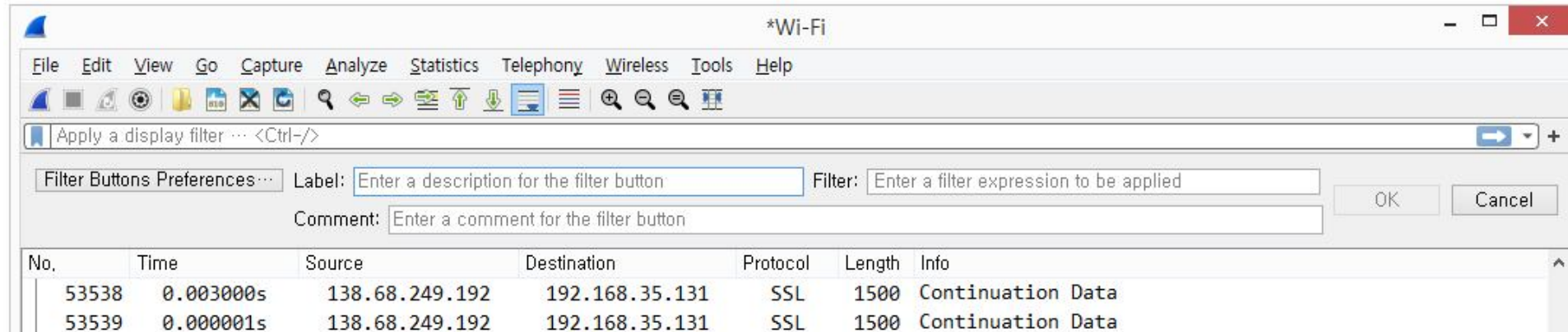
④ MAC 주소 기반의 트래픽 수집

- ether host 00:08:15:00:08:15
- ether src 00:08:15:00:08:15
- ether dst 00:08:15:00:08:15
- not ether host 00:08:15:00:08:15

특정 애플리케이션에 대한 트래픽 수집

- port 53
- not port 53
- port 80
- udp port 67
- tcp port 21
- portrange 1-80
- tcp portrange 1-80
- port 20 or port 21
- host 10.3.1.1 and port 80
- host 10.3.1.1 and not port 80
- udp src port 68 and udp dst port 67

2) 디스플레이 필터



적절한 디스플레이 필터 문법 사용

- 간단한 디스플레이 필터 문법

arp

ip

ipv6

tcp

적절한 디스플레이 필터 문법 사용

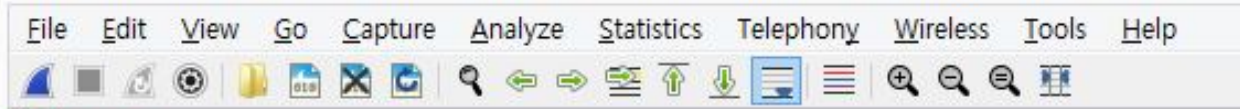
<<디스플레이 필터 오류 탐지 메커니즘 >>

- 대소문자 구분

- 적색 배경

- 문법 검사 실패

- 동작하지 않음



The image shows the Wireshark network protocol analyzer interface. The filter bar at the top contains the text 'http.re' and is highlighted with a red background. Below the filter bar, a table of captured packets is visible.

No.	Time	Source	Destination	Protocol
1	0.000000s	138.68.249.192	192.168.35.131	SSL
2	0.000069s	192.168.35.131	138.68.249.192	TCP

- 녹색 배경

- 문법 이상 없음

- ‘논리 검사’는 하지 않음 (예) http && udp

- 황색 배경

- 필터가 원하는 대로 동작하지 않는 것을 경고 (예) ip.addr != 10.1.1.1

디스플레이필터와 연산자 비교

연산자	영어표기	예제
==	eq	ip.src == 10.2.2.2
!=	ne	tcp.srcport != 80
>	gt	frame.time_relative > 1
<	lt	tcp.window_size < 1460
>=	ge	dns.count.answers >=10
<=	lt	ip.ttl < 10
	contains	http contain "GET"

캡처 필터 vs 디스플레이 필터

캡처 필터 구문 예제	디스플레이 필터 예제
host 172.16.1.1	ip.host == 172.16.1.1
src host 172.16.1.1	ip.src ==172.16.1.1
dst host 172.16.1.1	ip.dst ==172.16.1.1
port 8080	tcp.port == 8080
!port 8080	!tcp.port = 8080

❶ 단순 IP 주소 호스트에게/부터의 트래픽 필터링

- `ip.addr == 10.3.1.1`
- `!ip.addr == 10.3.1.1`
- `ipv6.addr == 2406:da00:ff00::6b16:f02d`
- `ip.src == 10.3.1.1`
- `ip.dst == 10.3.1.1`
- `ip.host == www.wireshark.org`

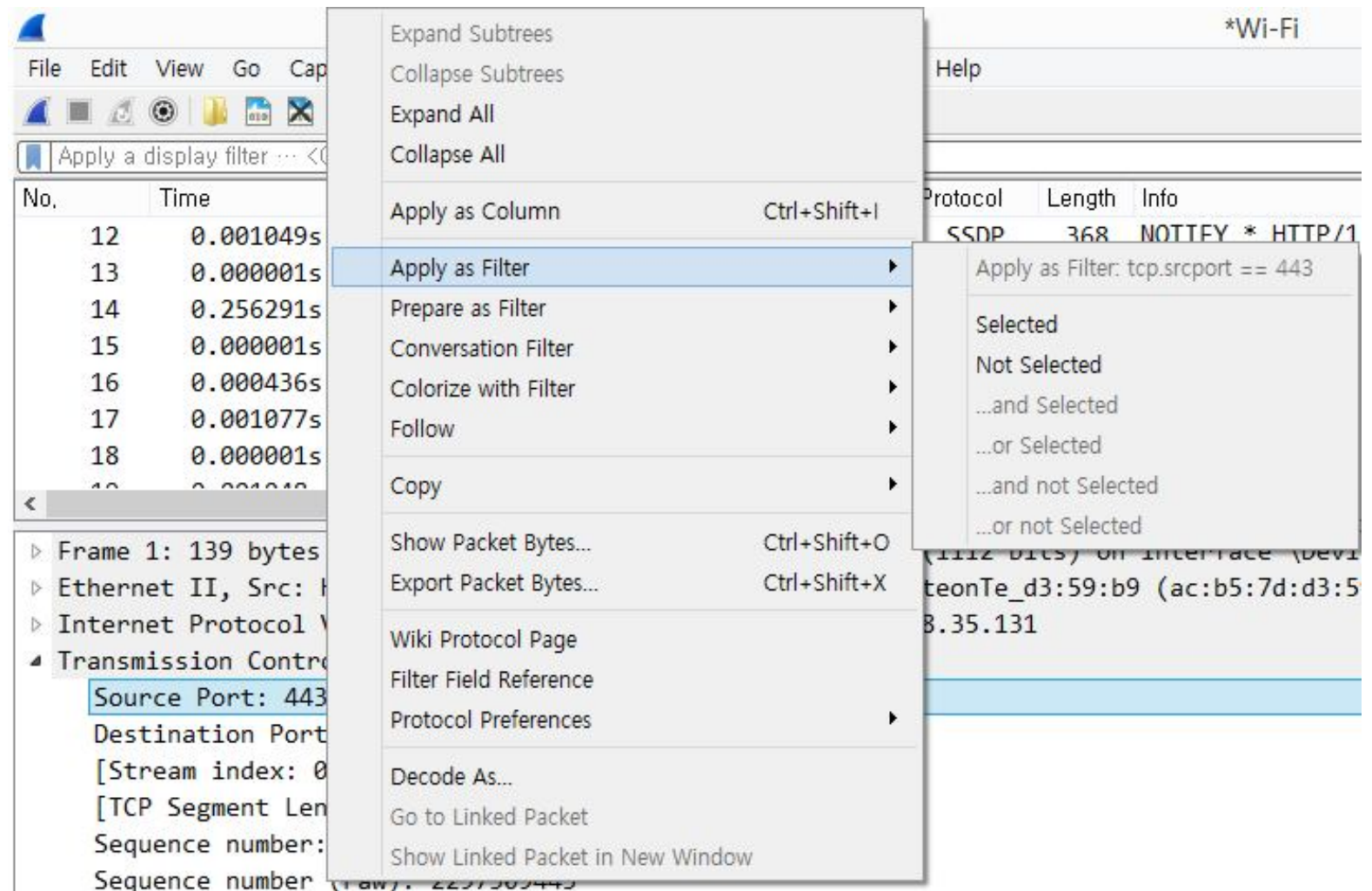
② 주소 범위에게/부터의 트래픽 필터링

- `ip.addr > 10.3.0.1 && ip.addr < 10.3.0.5`
- `(ip.addr >= 10.3.0.1 && ip.addr <= 10.3.0.6) && !ip.addr == 10.3.0.3`
- `ipv6.addr == fe80:: && ipv6.addr < fec0::`

③ IP 서브넷에서/으로부터 트래픽 필터링

- `ip.addr == 10.3.0.0/16`
- `ip.addr == 10.3.0.0/16 && !ip.addr == 10.3.0.3`
- `!ip.addr == 10.3.0.0/16 && !ip.addr == 10.2.0.0/16`

- Apply as Filter
- Prepare as Filter



Selected

$A == B$

Not Selected

$!(A == B)$

... and Selected

$(A == B) \&\& (C \dots)$

... or Selected

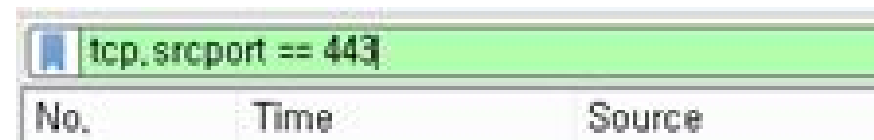
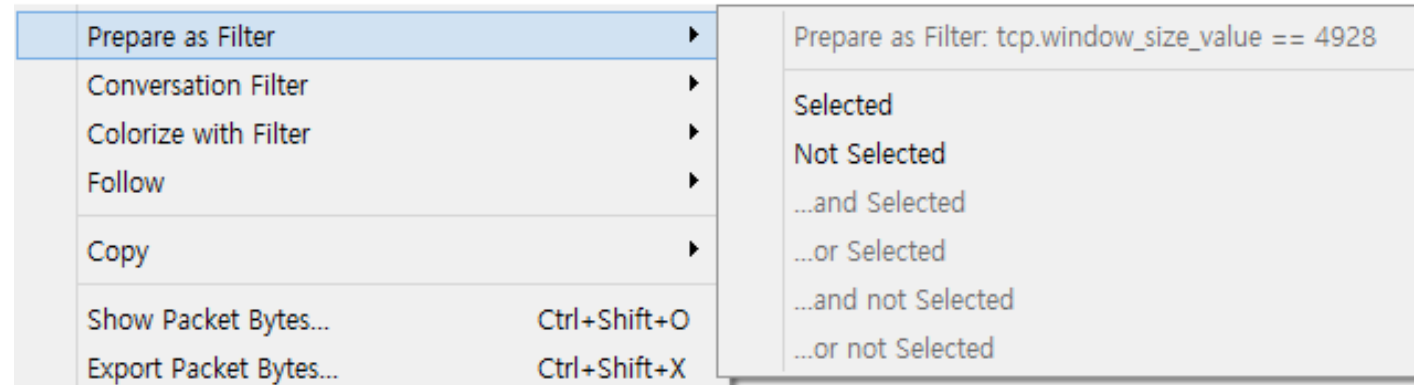
$(A == B) \parallel (C \dots)$

... and Not Selected

$(A == B) \&\& !(C \dots)$

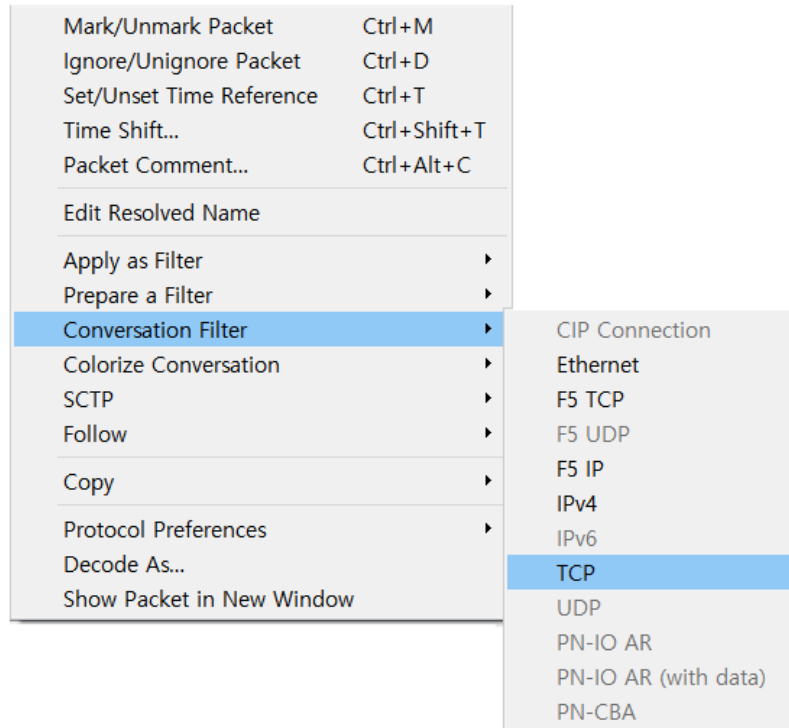
... or Not Selected

$(A == B) \parallel !(C \dots)$



3) Conversation Filter

❶ 패킷 리스트 > 패킷선택 > 오른쪽 마우스 클릭 > Conversation Filter > TCP



추적파일 : http-espn101.pcapng

(ip.addr eq 24.6.173.220 and ip.addr eq 199.181.132.250) and (tcp.port eq 19941 and tcp.port eq 80)						
No.	Time	Source	Destination	Protocol	Length	Info
5	0.000000	24.6.173.220	199.181.132.250	TCP	66	19941 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=
6	0.031335	199.181.132.250	24.6.173.220	TCP	66	80 → 19941 [SYN, ACK] Seq=0 Ack=1 Win=4380
7	0.000126	24.6.173.220	199.181.132.250	TCP	54	19941 → 80 [ACK] Seq=1 Ack=1 Win=65700 Len
8	0.000665	24.6.173.220	199.181.132.250	HTTP	603	GET / HTTP/1.1
9	0.041099	199.181.132.250	24.6.173.220	HTTP	484	HTTP/1.1 301 Moved Permanently (text/html
31	0.199860	24.6.173.220	199.181.132.250	TCP	54	19941 → 80 [ACK] Seq=550 Ack=431 Win=65268
4891	68.873340	24.6.173.220	199.181.132.250	TCP	54	19941 → 80 [RST, ACK] Seq=550 Ack=431 Win=

② Statistics > Conversation Filter

추적파일 : http-espn101.pcapng

Wireshark · Conversations · http-espn101.pcapng

Ethernet · 1 IPv4 · 37 IPv6 TCP · 63 UDP · 82											
Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
24.6.173.220	75.75.75.75	180	22 k	90	6973	90	15 k	0.000000	21.8143	2557	5526
24.6.173.220	199.181.132.250	7	1381	5	831	2	550	0.030245	69.1464	96	63
24.6.173.220	68.71.216.176	127	134 k	38	7147	89	127 k	0.168701	24.5121	2332	41 k
24.6.173.220	184.84.222.48	720	649 k	265	43 k	455	605 k	0.322923	70.0159	5024	69 k
24.6.173.220	143.127.102.125	10	1229	5	514	5	715	0.377829	0.1381	29 k	41 k
24.6.173.220	70.42.13.100	12	2578	7	1903	5	675	2.433476	14.8802	1023	362
24.6.173.220	68.71.212.151	7	1293	5	828	2	465	2.437970	66.7377	99	55
24.6.173.220	74.125.224.59	142	115 k	51	9643	91	105 k	2.843065	66.3320	1162	12 k
24.6.173.220	184.84.222.152	303	286 k	110	25 k	193	261 k	3.261301	70.9168	2865	29 k

Wireshark · Conversations · http-espn101.pcapng

Ethernet · 1 IPv4 · 37 IPv6 TCP · 63 UDP · 82											
Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
24.6.173.220	143.127.102.125	10	1229	5	514	5	715	0.377829	0.1381	29 k	41 k
24.6.173.220	68.71.212.151	7	1293								
24.6.173.220	199.181.132.250	7	1381								
24.6.173.220	107.20.148.253	10	1400								
24.6.173.220	184.84.222.64	8	1422								
24.6.173.220	184.51.159.181	10	1437	6	888						
24.6.173.220	184.84.183.147	8	1573	5	699						
24.6.173.220	107.22.175.32	10	1602	6	1068	4	55				
24.6.173.220	184.84.222.112	8	2120	5	701	3	141				
24.6.173.220	64.95.73.7	9	2311	6	1304	3	100				
24.6.173.220	68.71.220.175	7	2355	5	1171	2	118				

② Statistics > Conversation Filter

- Packets 필드를 기준으로 내림 차순으로 정렬
- 첫 번째 패킷 선택
- 오른쪽 마우스 클릭 > Apply as Filter > Selected > A → B

Wireshark · Conversations · http-espn101.pcapng

Ethernet · 1 IPv4 · 37 IPv6 TCP · 63 UDP · 82										
Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B
24.6.173.220	184.84.222.88	1,855	2020 k	533						3765
24.6.173.220	184.84.222.48	720	649 k	265						5024
24.6.173.220	184.84.222.120	613	628 k	195						1882
24.6.173.220	184.84.222.10	371	382 k	119						980
24.6.173.220	184.84.222.152	303	286 k	110						2865
24.6.173.220	75.75.75.75	180	22 k	90	6973	90				2557
24.6.173.220	74.125.224.59	142	115 k	51	9643	91				1162
24.6.173.220	68.71.216.157	132	20 k	66	3672	66	16 k	21.802866	4	658
24.6.173.220	68.71.216.176	127	134 k	38	7147	89	127 k	0.168701	2	2332
24.6.173.220	184.84.222.16	41	36 k	15	1768	26	35 k	7.951909	6	231
24.6.173.220	50.17.254.18	37	7626	22	5637	15	1989	9.581595	2.8209	15 k
24.6.173.220	184.84.222.75	36	33 k	12	1602	24	32 k	5.377013	63.7964	200

[참고] Endpoints

Statistics > Endpoints

Wireshark · Endpoints · http-espn101.pcapng

Ethernet · 2 IPv4 · 38 IPv6 TCP · 100 UDP · 83

Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	AS Number	AS Organization
24.6.173.220	4,900	4490 k	1,700	201 k	3,200	4288 k	—	—	—	—
184.84.222.88	1,855	2020 k	1,322	1989 k	533	30 k	—	—	—	—
184.84.222.48	720	649 k	455	605 k	265	43 k	—	—	—	—
184.84.222.120	613	628 k	418	614 k	195	14 k	—	—	—	—
184.84.222.10	371	382 k	252	374 k	119	7502	—	—	—	—
184.84.222.152	303	286 k	193	261 k	110	25 k	—	—	—	—
75.75.75.75	180	22 k	90	15 k	90	6973	—	—	—	—
74.125.224.59	142	115 k	91	105 k	51	9643	—	—	—	—
68.71.216.157	132	20 k	66	16 k	66	3672	—	—	—	—
68.71.216.176	127	134 k	89	127 k	38	7147	—	—	—	—
184.84.222.16	41	36 k	26	35 k	15	1768	—	—	—	—
50.17.254.18	37	7626	15	1989	22	5637	—	—	—	—
184.84.222.75	36	33 k	24	32 k	12	1602	—	—	—	—
138.108.7.20	31	24 k	20	23 k	11	1675	—	—	—	—
184.84.222.137	30	19 k	16	17 k	14	1638	—	—	—	—
68.71.216.171	29	24 k	17	23 k	12	1007	—	—	—	—
96.17.110.92	25	8423	12	4121	13	4302	—	—	—	—
96.17.148.114	24	13 k	12	10 k	12	3575	—	—	—	—
96.17.110.102	17	2650	6	672	11	1978	—	—	—	—
66.235.138.59	15	7482	7	2816	8	4666	—	—	—	—

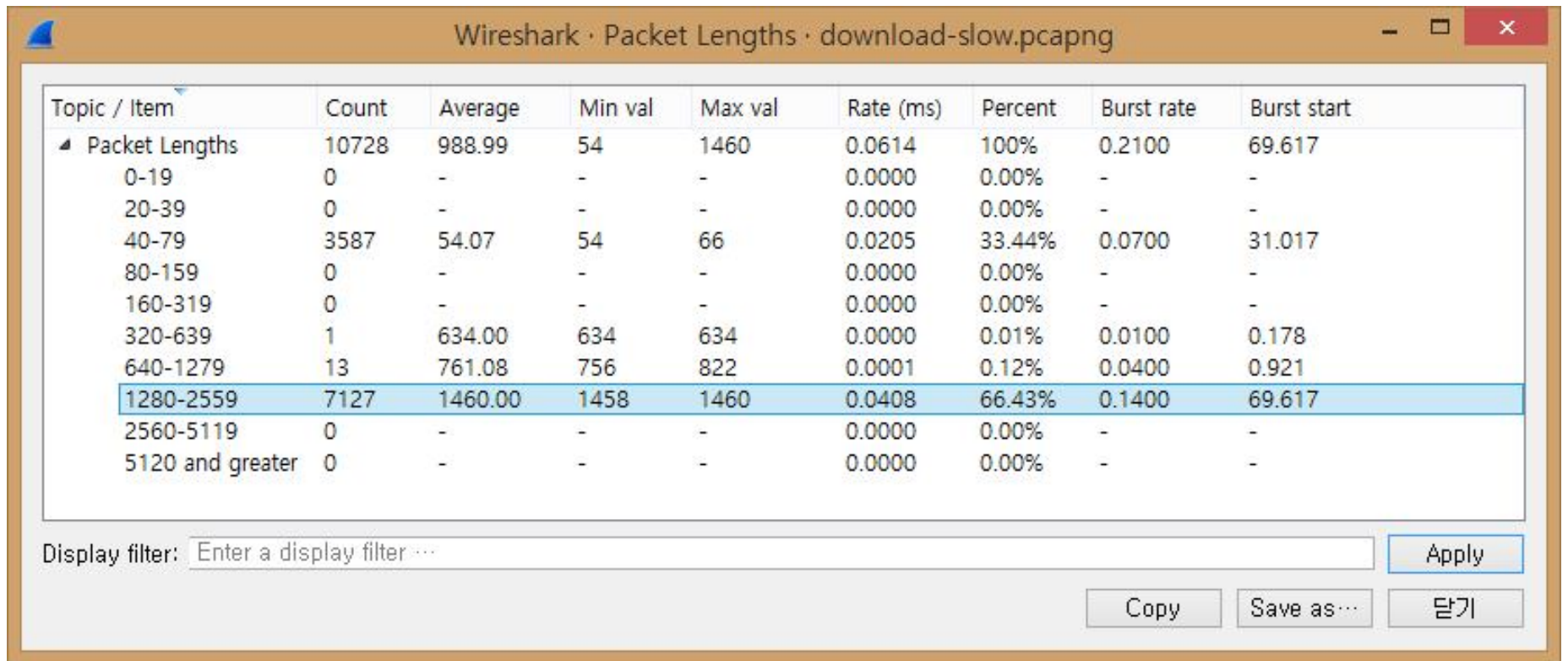
☐ Name resolution ☐ Limit to display filter

Endpoint Types

Copy Map 달기 도움말

* Statistics > Packet Lengths

- 대용량 캡처 파일의 구성을 이해하고자 할 때 유용
- 길이에 따라 패킷 분포를 확인



The screenshot shows the 'Wireshark · Packet Lengths · download-slow.pcapng' window. It displays a table of packet length statistics. The '1280-2559' range is highlighted in blue, indicating it is the selected item. Below the table, there is a 'Display filter' input field and buttons for 'Apply', 'Copy', 'Save as...', and '닫기' (Close).

Topic / Item	Count	Average	Min val	Max val	Rate (ms)	Percent	Burst rate	Burst start
Packet Lengths	10728	988.99	54	1460	0.0614	100%	0.2100	69.617
0-19	0	-	-	-	0.0000	0.00%	-	-
20-39	0	-	-	-	0.0000	0.00%	-	-
40-79	3587	54.07	54	66	0.0205	33.44%	0.0700	31.017
80-159	0	-	-	-	0.0000	0.00%	-	-
160-319	0	-	-	-	0.0000	0.00%	-	-
320-639	1	634.00	634	634	0.0000	0.01%	0.0100	0.178
640-1279	13	761.08	756	822	0.0001	0.12%	0.0400	0.921
1280-2559	7127	1460.00	1458	1460	0.0408	66.43%	0.1400	69.617
2560-5119	0	-	-	-	0.0000	0.00%	-	-
5120 and greater	0	-	-	-	0.0000	0.00%	-	-

- 길이가 긴 패킷이 66.43% 차지, 짧은 길의 패킷은 33.44%

*Statistics > Protocol Hierarchy

Wireshark - Protocol Hierarchy Statistics - http_espn.pcapng

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s
▼ Frame	100.0	956	100.0	652181	2548 k	0	0	0
▼ Ethernet	100.0	956	2.1	13384	52 k	0	0	0
▼ Internet Protocol Version 4	100.0	956	2.9	19120	74 k	0	0	0
▼ User Datagram Protocol	2.9	28	0.0	224	875	0	0	0
Domain Name System	2.9	28	0.3	2017	7880	28	2017	7880
▼ Transmission Control Protocol	97.1	928	94.7	617436	2412 k	426	14032	54 k
▼ Hypertext Transfer Protocol	52.5	502	90.1	587338	2294 k	441	517225	2020 k
Unreassembled Fragmented Packet	0.1	1	0.0	0	0	1	0	0
▼ Portable Network Graphics	1.5	14	1.3	8607	33 k	8	3064	11 k
Unreassembled Fragmented Packet	0.6	6	0.0	0	0	6	0	0
Media Type	0.3	3	0.4	2741	10 k	3	2741	10 k
Line-based text data	1.5	14	2.3	15259	59 k	14	15259	59 k
▼ JPEG File Interchange Format	1.9	18	2.5	16201	63 k	13	11575	45 k
Unreassembled Fragmented Packet	0.5	5	0.0	0	0	5	0	0
eXtensible Markup Language	0.2	2	0.9	5868	22 k	2	5868	22 k
▼ Compuserve GIF	0.9	9	0.8	4946	19 k	5	1254	4899
Unreassembled Fragmented Packet	0.4	4	0.0	0	0	4	0	0

• Statistics > Conversation

Wireshark - Conversations - http_espn_fail.pcapng

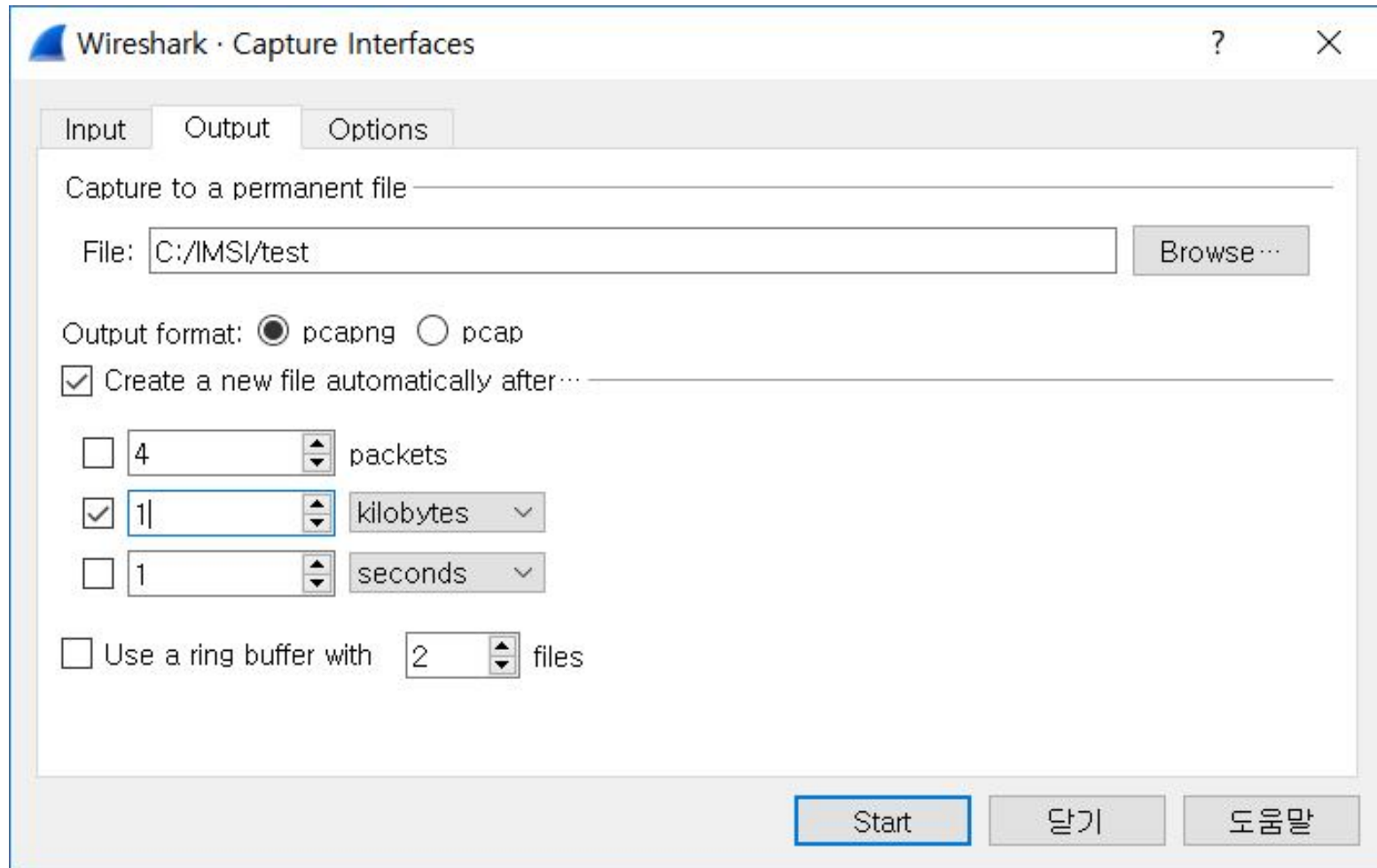
Ethernet · 1 IPv4 · 8 IPv6 TCP · 16 UDP · 7											
Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
172.16.16.154	4.2.2.1	14	1627	7	521	7	1106	0.000000	0.6639	6278	
172.16.16.154	68.71.212.158	13	2032	7	832	6	1200	0.027167	90.8752	73	
172.16.16.154	199.181.133.61	61	49 k	24	1953	37	47 k	0.238547	91.0836	171	
172.16.16.154	203.0.113.94	93	6774	93	6774	0	0	0.430071	94.5936	572	
172.16.16.154	72.21.91.8	92	70 k	43	3170	49	67 k	0.526867	60.5532	418	
172.16.16.154	72.246.56.35	247	196 k	113	8315	134	188 k	0.527902	90.8063	732	
172.16.16.154	69.31.75.194	19	9949	9	1007	10	8942	0.579477	90.6593	88	
172.16.16.154	72.246.56.83	30	20 k	15	1518	15	19 k	0.659868	45.3449	267	

Tshark을 이용한 패킷 수집

	LINUX	MS-Window
패킷수집	TCPdump	TShark
패킷분석	Wireshark	Wireshark

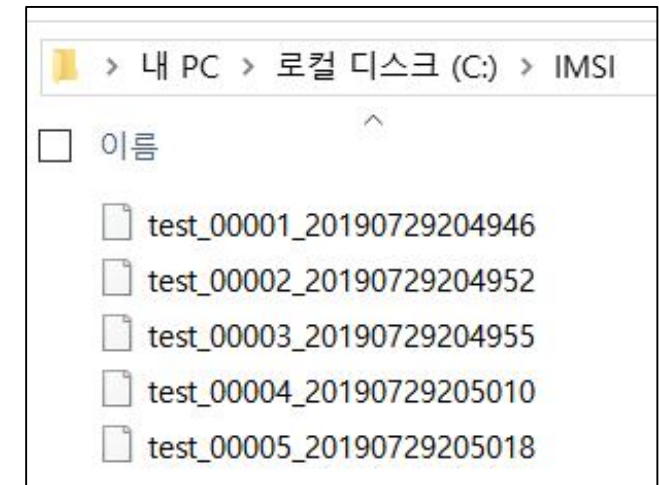
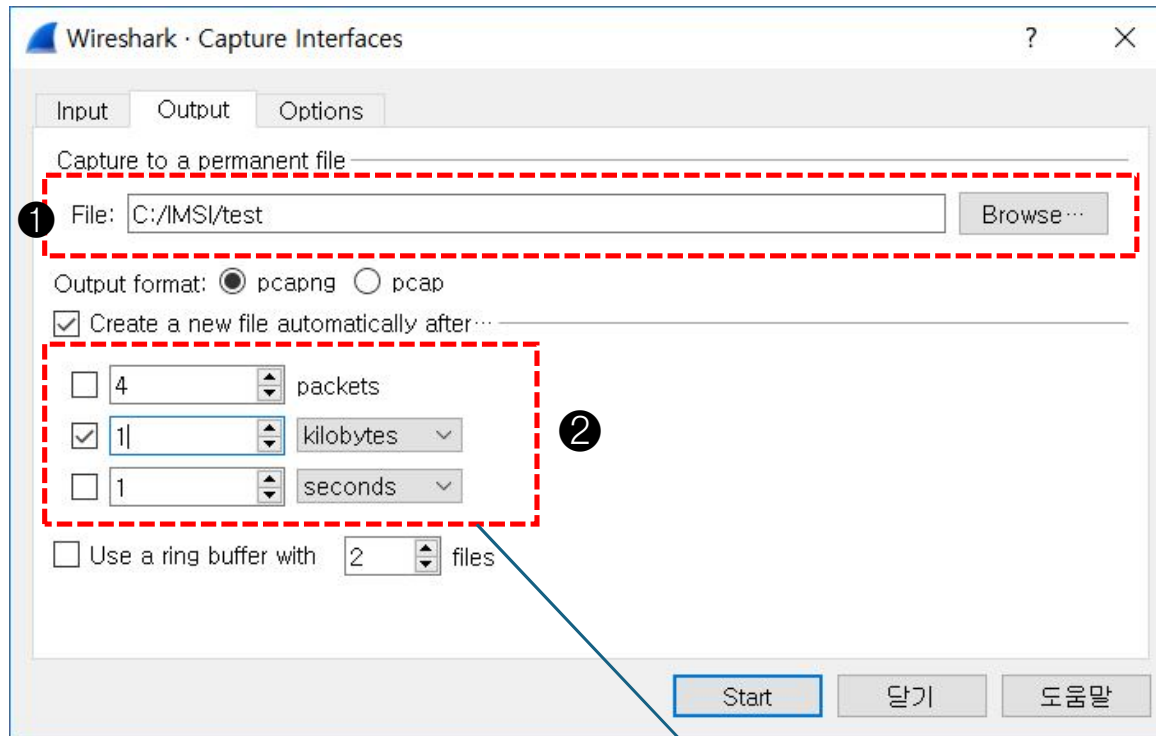
파일 집합으로 수집

- Capture Options > Output Tab > Create a new file automatically after...



파일 집합으로 수집

- Capture Options > Output Tab > Create a new file automatically after...



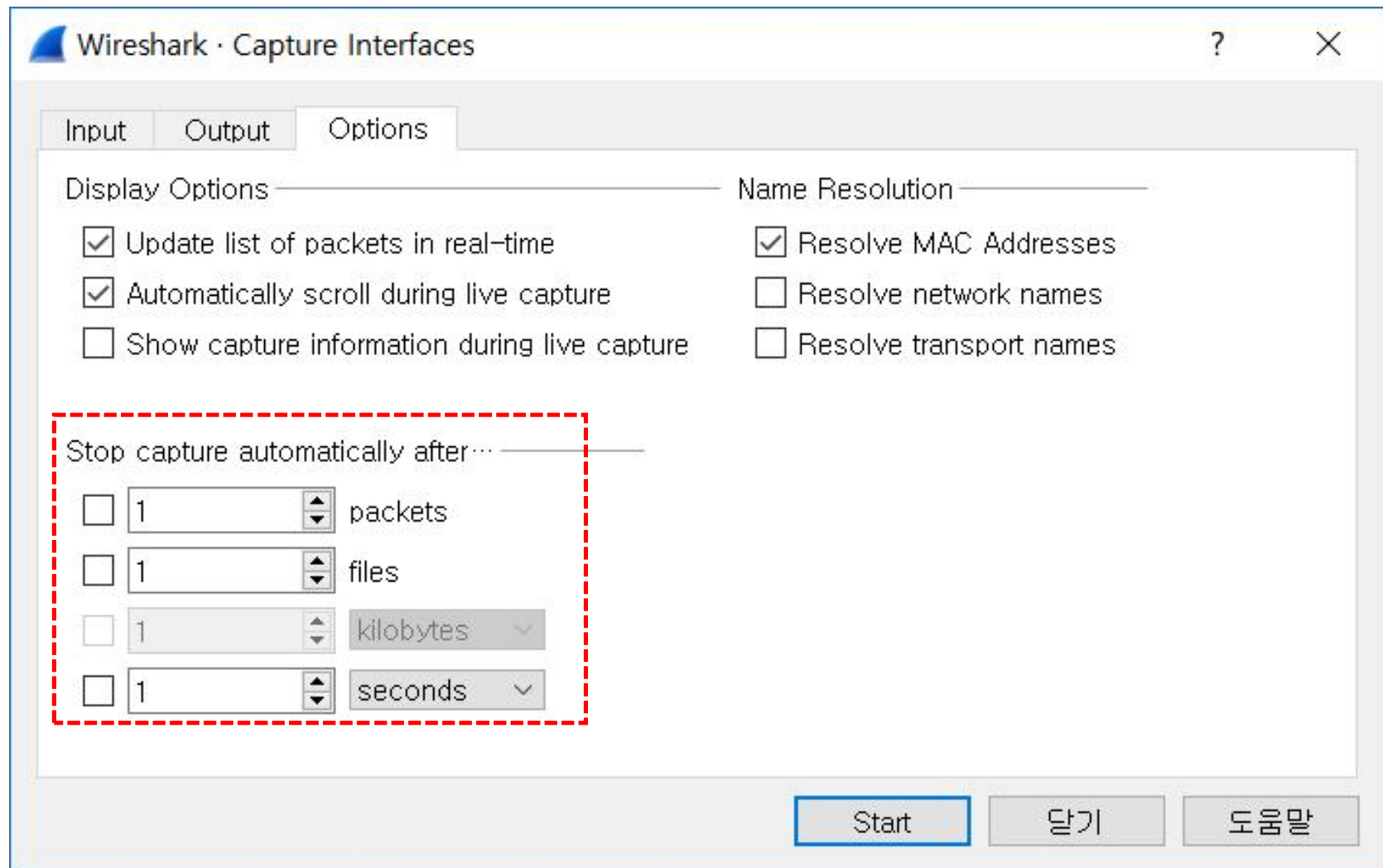
- File > Open
 - File > File set > List Files

파일당 4개의 패킷
1MB 파일 크기
1초마다

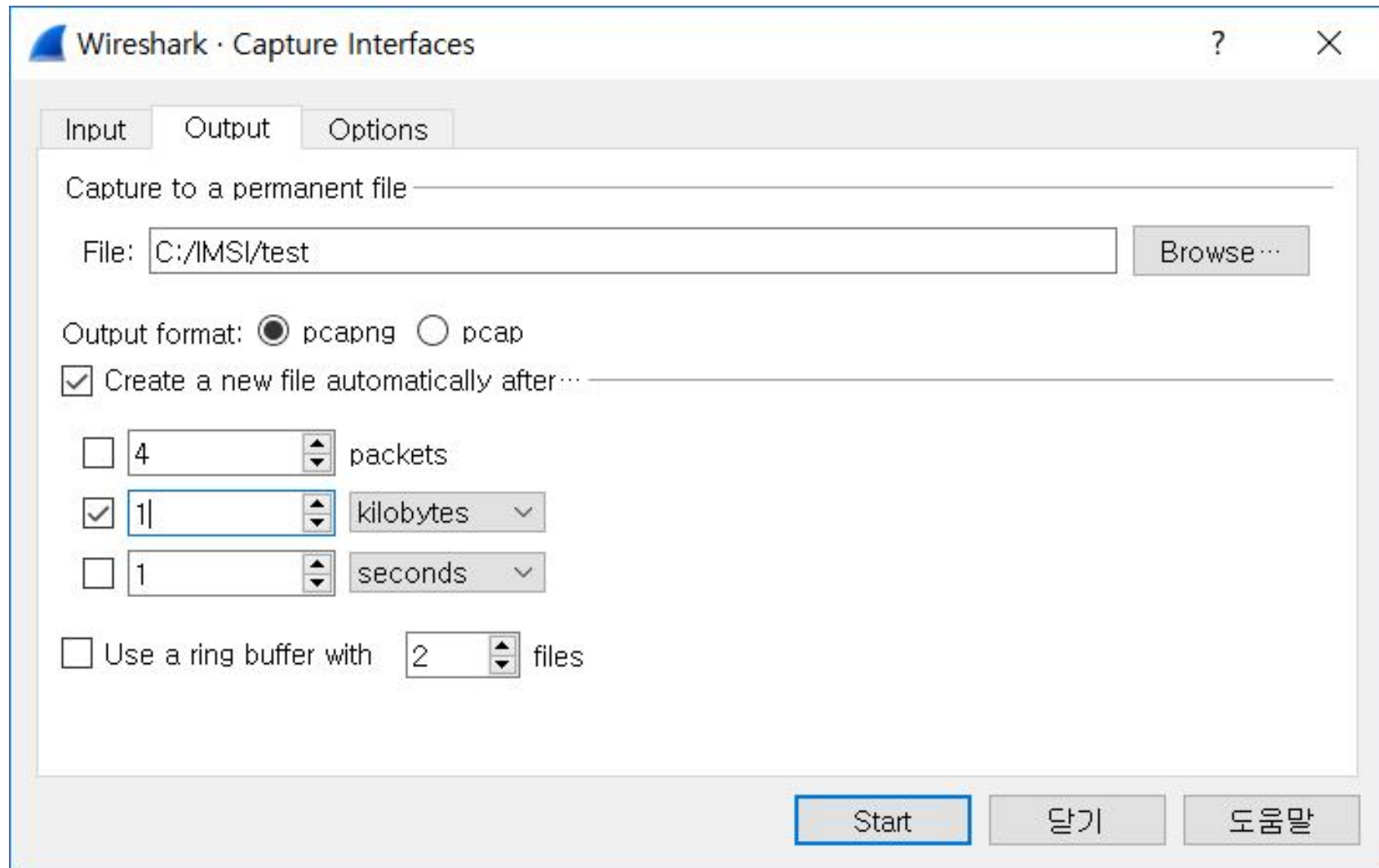
위의 어느 조건이든
먼저 만나면 파일 생성

파일 집합으로 수집

- Capture Options > Option Tab > Stop capture automatically after....

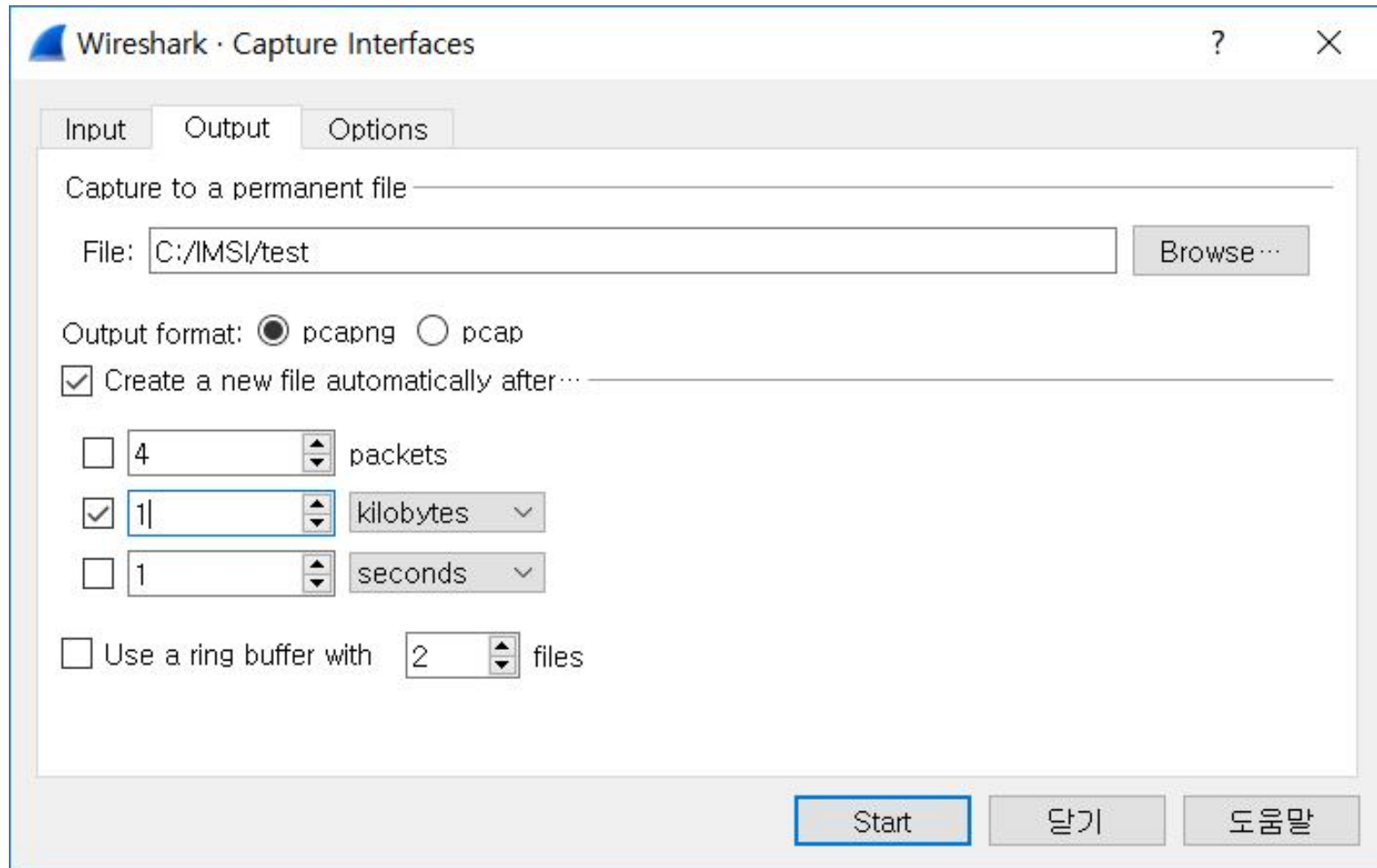


링 버퍼 사용



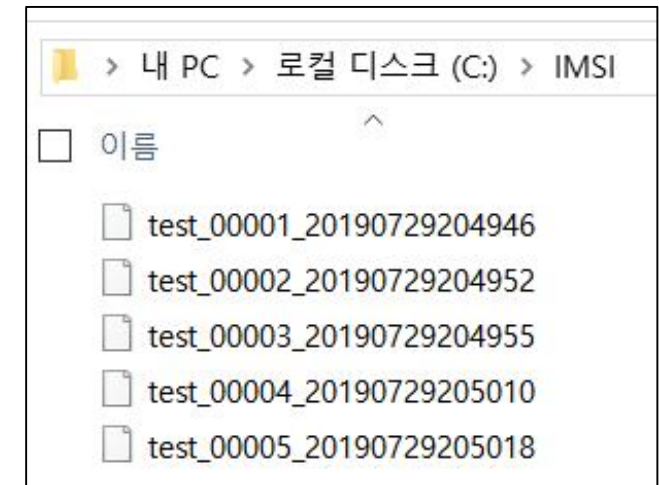
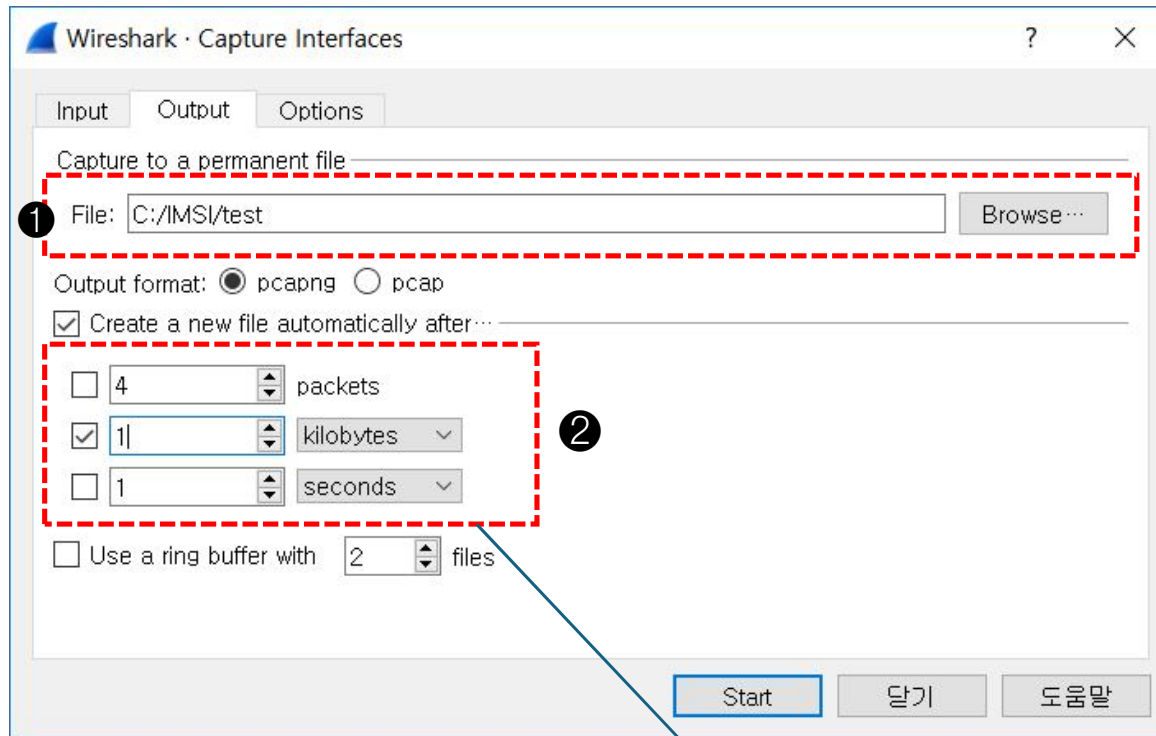
파일 집합으로 수집

- Capture Options > Output Tab > Create a new file automatically after...



파일 집합으로 수집

- Capture Options > Output Tab > Create a new file automatically after...



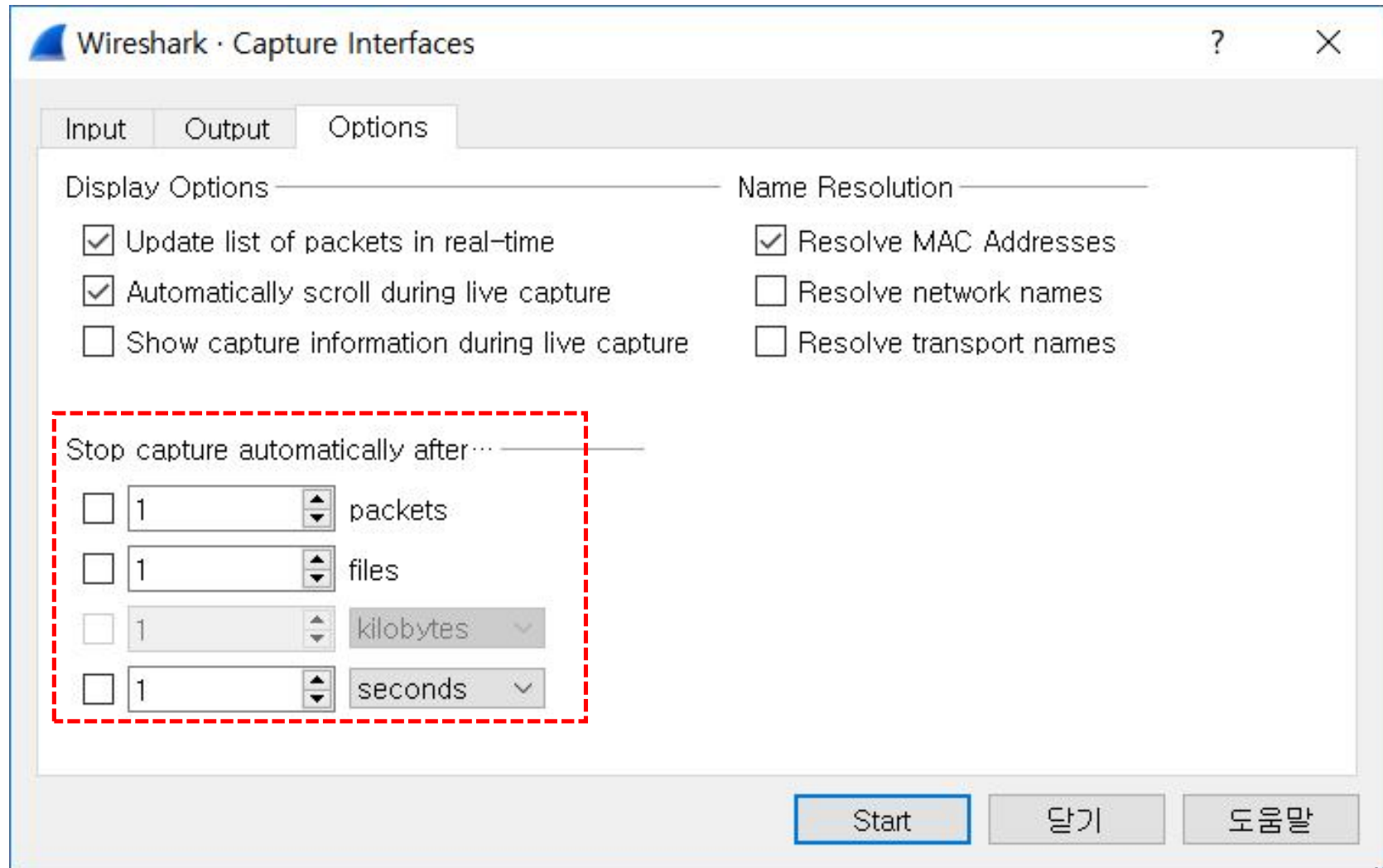
- File > Open
 - File > File set > List Files

파일당 4개의 패킷
1MB 파일 크기
1초마다

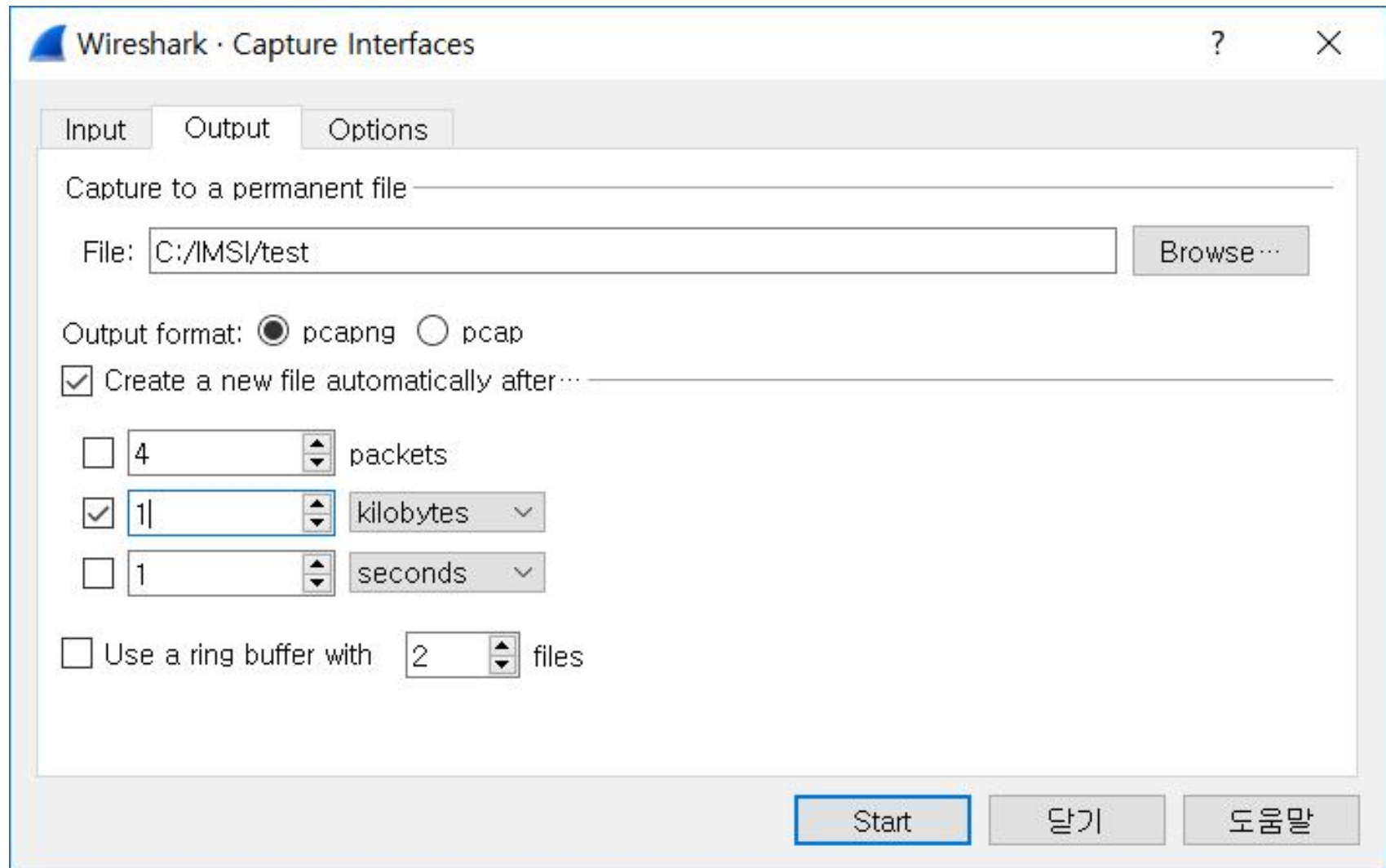
위의 어느 조건이든
먼저 만나면 파일 생성

파일 집합으로 수집

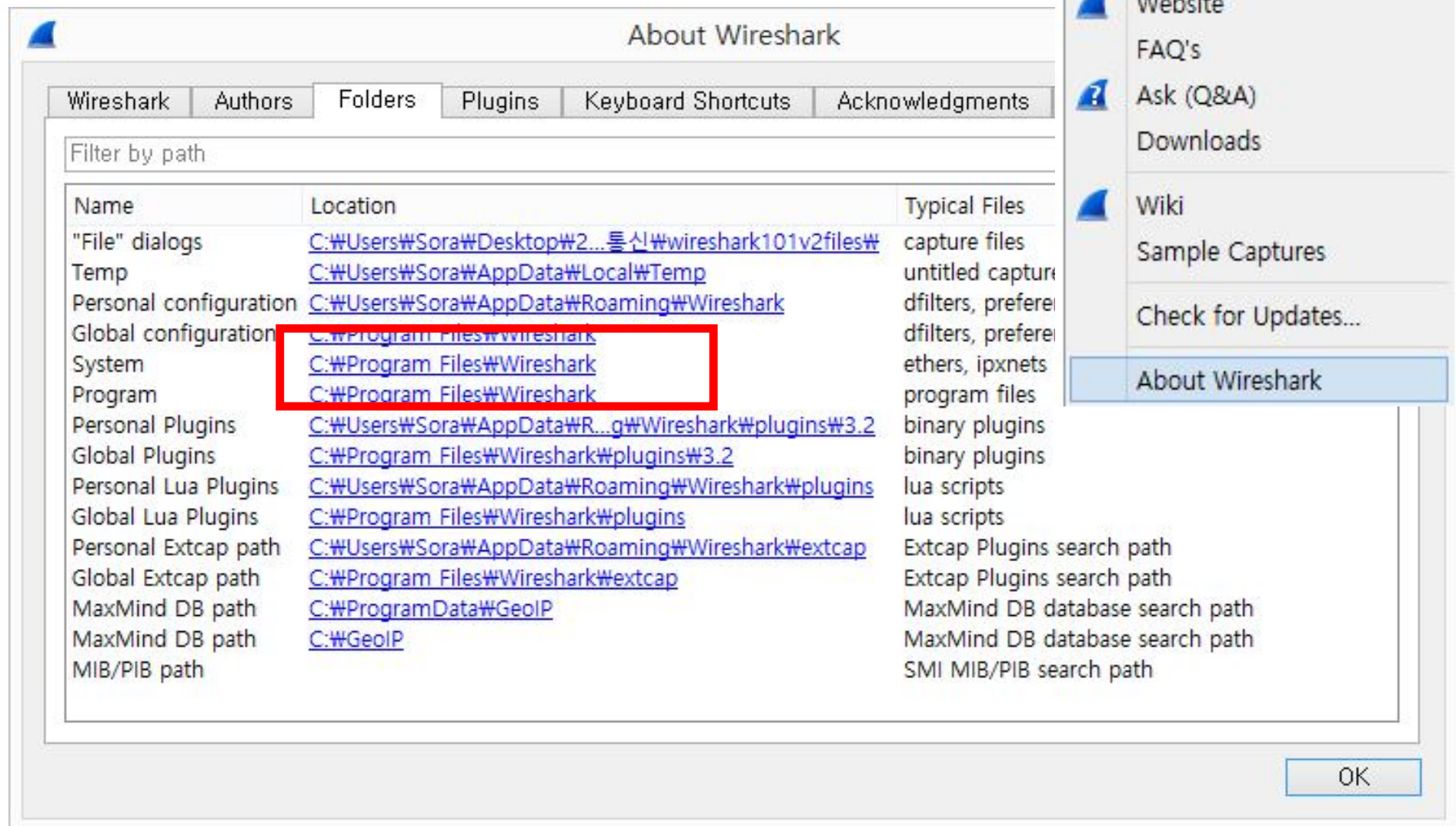
- Capture Options > Option Tab > Stop capture automatically after....



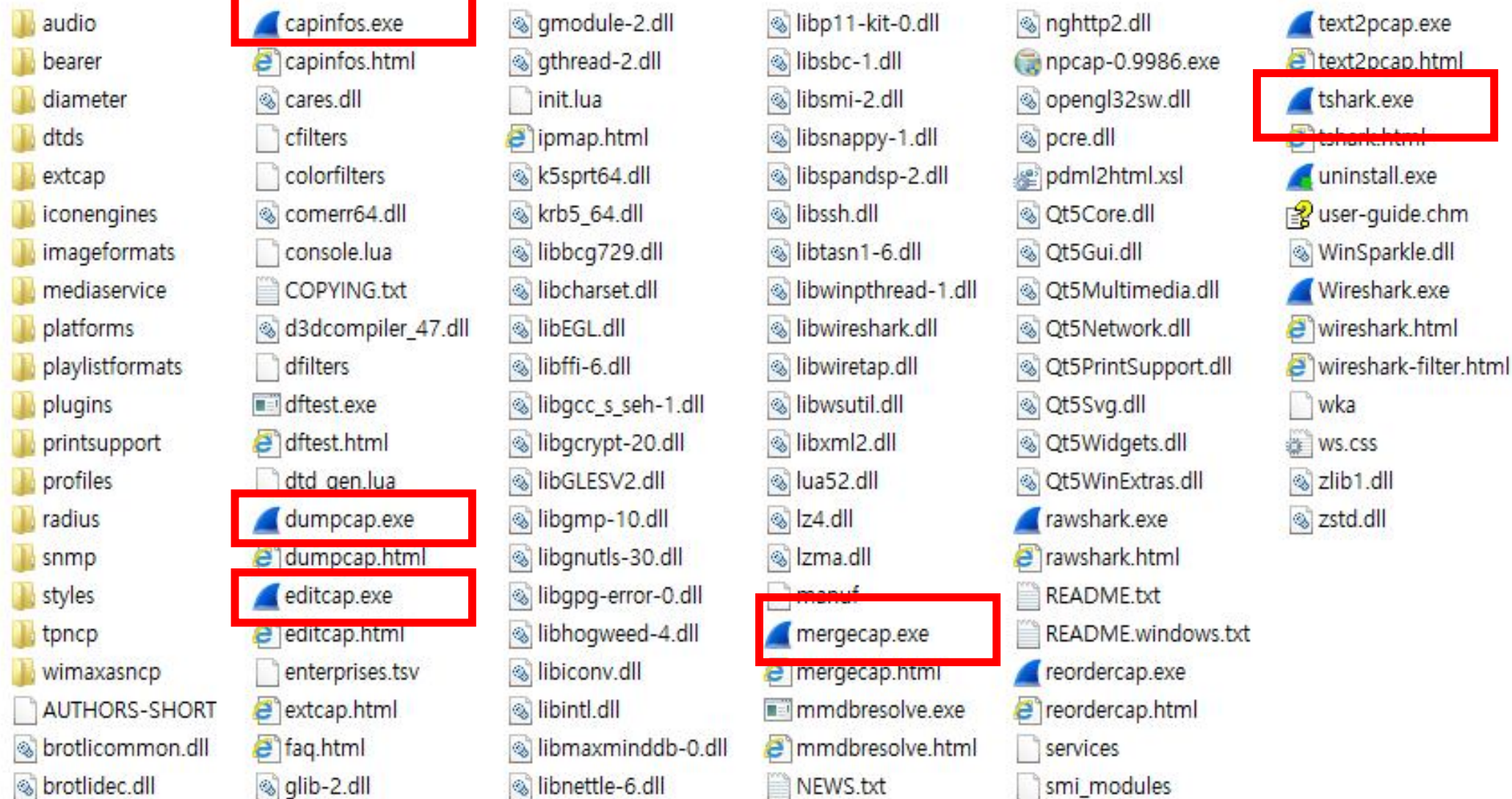
링 버퍼 사용



Tshark을 이용한 패킷 수집



내 PC > 로컬 디스크 (C:) > Program Files > Wireshark >



커맨드라인에서 트래픽 수집

- dumpcap.exe나 tshark.exe를 이용해 커맨드 라인으로 트래픽 수집
 - Tshark를 구동하면 dumpcap.exe를 호출해서 수집 기능을 활용

```
C:\Program Files\Wireshark>dir dumpcap.exe
```

```
C:\Program Files\Wireshark 디렉터리
```

```
2020-02-27 오전 05:47          420,416 dumpcap.exe
                  1개 파일          420,416 바이트
                  0개 디렉터리 86,740,180,992 바이트 남음
```

```
C:\Program Files\Wireshark>dir tshark.exe
```

```
C:\Program Files\Wireshark 디렉터리
```

```
2020-02-27 오전 05:47          582,720 tshark.exe
                  1개 파일          582,720 바이트
                  0개 디렉터리 86,740,180,992 바이트 남음
```

① tshark -h

```
C:\Program Files\Wireshark>tshark -h
TShark (Wireshark) 3.0.3 (v3.0.3-0-g6130b92b0ec6)
Dump and analyze network traffic.
See https://www.wireshark.org for more information.

Usage: tshark [options] ...
```

② tshark -D

```
C:\Program Files\Wireshark>tshark -D
1. \Device\NPF_{5256CF9B-1707-460C-B397-669CE852CFDE}
2. \Device\NPF_{A378EED4-AC87-4168-8B22-DFDD0B9EC62E}
3. \Device\NPF_{557124E1-28F6-4C2F-BDCE-1DFD75D011CE} (Wi-Fi)
4. \Device\NPF_{630C09BF-1CAB-457F-978E-7A0BBEE76CF5}
5. \Device\NPF_{E8E017EF-FC6D-4933-BD98-9AEC5CB26CF6}
6. \Device\NPF_{27056470-1B11-4C61-8E92-0B25B7CB57C9}
7. \Device\NPF_{DFBA2BCB-8989-40B4-873C-8CF862935F18}
8. \Device\NPF_{6EC32DD5-41B3-45FF-B701-BB15DC1E7E45}
9. \Device\NPF_{23E2556A-B636-483F-A13E-C3E889DD3825}
10. \Device\NPF_{415D90F5-DFA5-426C-ABFE-BCCAB1542370}
```

③ tshark -i 3

```
C:\Program Files\Wireshark>tshark -i 3
Capturing on 'Wi-Fi'
 1  0.000000 192.168.35.145 → 69.167.144.15 66 65160 64020 → https(443) [SYN] Seq=0 Win=65160 Len=0 MSS=1460 WS=256 SACK_PERM=1
 2  3.816065 192.168.35.145 → 121.53.202.218 66 65535 64021 → https(443) [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
 3  3.816070 192.168.35.145 → 121.53.202.218 66 65535 64022 → https(443) [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
 4  3.818776 121.53.202.218 → 192.168.35.145 66 8190 https(443) → 64022 [SYN, ACK] Seq=0 Ack=1 Win=8190 Len=0 MSS=1460 WS=16 SACK_PERM=1
 5  3.818895 192.168.35.145 → 121.53.202.218 54 1024 64022 → https(443) [ACK] Seq=1 Ack=1 Win=262144 Len=0
 6  3.819200 121.53.202.218 → 192.168.35.145 66 8190 https(443) → 64021 [SYN, ACK] Seq=0 Ack=1 Win=8190 Len=0 MSS=1460 WS=16 SACK_PERM=1
 7  3.819311 192.168.35.145 → 121.53.202.218 54 1024 64021 → https(443) [ACK] Seq=1 Ack=1 Win=262144 Len=0
 8  3.819713 192.168.35.145 → 121.53.202.218 295 1024 Client Hello
 9  3.819864 192.168.35.145 → 121.53.202.218 295 1024 Client Hello
```

* Ctrl+C 로 중단

④ tshark -i 3 -w TST.pcapng

```
C:\Program Files\Wireshark>tshark -i 3 -w TST.pcapng
Capturing on 'Wi-Fi'
2472
```

```
C:\Program Files\Wireshark>
```

⑤ tshark -r TST.pcapng

```
C:\Program Files\Wireshark>tshark -r TST.pcapng
 1  0.000000 192.168.35.145 → 69.167.144.15 74 65160 64159 → ht
 2  0.749769 192.168.35.145 → 104.74.232.184 54 256 64153 → ht
 3  0.752502 104.74.232.184 → 192.168.35.145 54 237 http(80) →
 4  0.752540 192.168.35.145 → 104.74.232.184 54 256 64153 → ht
 5  1.055385 192.168.35.145 → 110.76.141.124 66 64240 64160 → h
 6  1.058892 110.76.141.124 → 192.168.35.145 66 29200 https(443)
 7  1.058938 192.168.35.145 → 110.76.141.124 54 256 64160 → ht
 8  1.059047 192.168.35.145 → 110.76.141.124 256 256 Client Hel
```

⑥ tshark -i 3 -a files:3 -b duration:10 -w myshark.pcapng

```
C:\Program Files\Wireshark>tshark -i 3 -a files:3 -b duration:10 -w myshark.pcapng
Capturing on 'Wi-Fi'
4464
```

```
C:\Program Files\Wireshark>dir myshark*
C 드라이브의 볼륨에는 이름이 없습니다.
볼륨 일련 번호: BCF8-FA59
```

```
C:\Program Files\Wireshark 디렉터리
```

```
2018-08-12 오전 12:46          679,312 myshark_00001_20190801004608.pcapng
2018-08-12 오전 12:46      1,269,276 myshark_00002_20190801004618.pcapng
2018-08-12 오전 12:46      937,040 myshark_00003_20190801004629.pcapng
                3개 파일              2,885,628 바이트
                0개 디렉터리 126,970,032,128 바이트 남음
```

-a file:3 3개 파일 수집 후 자동 정지

-b duration:10 10초후에 다음 파일을 생성

-w myshark.pcapng 추적파일명

커맨드라인 수집 과정에서 수집 필터(캡처필터)

① `tshark -i 3 -f "tcp port 443" -w mysecport443.pcapng`

```
C:\Program Files\Wireshark>tshark -i 3 -f "tcp port 443" -w mysecport443.pcapng
Capturing on 'Wi-Fi'
2734
```

```
C:\Program Files\Wireshark>tshark -r mysecport443.pcapng
 1  0.000000 192.168.35.145 → 69.167.144.15 74 65160 64805 → https(443) [SYN] Seq=0 W
 2  3.010265 192.168.35.145 → 69.167.144.15 74 65160 [TCP Retransmission] 64805 → htt
 3  3.627403 192.168.35.145 → 52.175.23.79 66 64240 64806 → https(443) [SYN] Seq=0 Wi
 4  3.710985 52.175.23.79 → 192.168.35.145 66 8192 https(443) → 64806 [SYN, ACK] Seq=
 5  3.711076 192.168.35.145 → 52.175.23.79 54 258 64806 → https(443) [ACK] Seq=1 Ack=
 6  3.711563 192.168.35.145 → 52.175.23.79 274 258 Client Hello
 7  3.792328 52.175.23.79 → 192.168.35.145 1514 1026 [TCP segment of a reassembled PDU
 8  3.792330 52.175.23.79 → 192.168.35.145 1514 1026 https(443) → 64806 [ACK] Seq=146
 9  3.792335 52.175.23.79 → 192.168.35.145 538 1026 Server Hello, Certificate, Server
```

② `tshark -i 3 -f "tcp port 443 and host 192.168.1.1" -w my443.pcapng`

커맨드라인 수집 과정에서 디스플레이 필터

❶ **tshark -r “mysecport443.pcapng” -Y “tcp.analysis.flags”**

```
C:\Program Files\Wireshark>tshark -r "mysecport443.pcapng" -Y "tcp.analysis.flags"
  2    3.010265 192.168.35.145 → 69.167.144.15 74 65160 [TCP Retransmission] 64805 → https(443)
1113   9.019700 192.168.35.145 → 69.167.144.15 66 65160 [TCP Retransmission] 64805 → https(443)
2734  24.038230 192.168.35.145 → 69.167.144.15 74 65160 [TCP Retransmission] 64898 → https(443)
```

❷ **tshark -r “mysecport443.pcapng” -Y “tcp.analysis.flags” -w tcpflag.pcapng**

❸ **tshark -r “mysecport443.pcapng” -Y “http.request.method == GET”**